

# The surreal and surcomplex reports

Fifty-one mathematical research packages,  
on the surreal field, its algebraic closure, its quaternion algebra,  
their reach into physics, and the foundations and computation of all of them

September 2026

## Scope

This document catalogues the fifty-one research packages stored under `docs`. Each entry names the problem the report attacks, the result it claims, and the delivery it arrived in.

The collection divides in five, and so does the directory tree. Eighteen reports, under `docs/surreal`, are about the surreal field  $\mathbf{No}$  itself and its real Hahn subfields: birthdays and their behaviour under multiplication, the option graphs of surreal forms, genetic functions and gaps, two questions of Lipparini about evaluating formal sums, the rigidity of exponential automorphisms, Gamma functions, differential transcendence of explicit surreals, Hahn-valued measures and probability, Markov generators, transcendence over bounded-support series, diagonal matrix scaling, vector and tensor fields, Euclidean three-space and its rotations, and finite probability, and the arithmetic and set-sized quotients of omnific integers. Twenty-six, under `docs/surcomplex`, are about the algebraic closure  $\mathbf{No}[i]$ : the theory of holomorphic functions over it, its differential algebra, its rank-one analytic geometry, its finite and infinite-dimensional spectral theory, its local, quasi-periodic and expanding dynamics, its nonabelian support obstructions, its entire functions at arbitrary valuation rank and their holonomic rigidity, Tate curves and multiscale theta series, Herglotz positivity, moment reconstruction, the summability of Wick expansions, the duals of Hahn vector spaces, the automorphisms of the field, cardinally bounded Hahn fields, what one dilation defines, the Gamma and zeta functions, and hidden negative directions of Hermitian forms. One, under `docs/surquaternions`, leaves commutativity behind for Hamilton's division algebra  $\mathbb{H}_{\mathbf{No}}$  over  $\mathbf{No}$ . One, under `docs/physics`, is the only report here whose subject is not mathematics: it asks what surreal scalars do and do not buy in general relativity, and answers negatively. The last five, under `docs/foundations-and-computation`, stand outside all of these: they are about the subject rather than inside it, asking in what foundation this mathematics legally lives, how one computes with it exactly, which surreal numbers are computable at all, how the surreal field of one set-theoretic universe looks from a larger one, and which universe of hereditary sets a birthday cutoff remembers.

These reports were previously part of a larger collection, in a repository whose subject was elsewhere; they were moved here with their git history. That is why an entry may refer to a merge, a duplicate sweep or a delivery number: those describe how the report reached its present form, and the record is kept rather than rewritten.

Each directory holds a  $\text{\LaTeX}$  source and a `README.md`; all except the newly placed set-sized-quotient report also have a current typeset article. Most include verification code and its recorded output. The omnific reports are partially integrated or still placed bases, as their entries specify. Source manuscripts for merged reports remain in Git history; their provenance notes record contributions and editorial choices. These are AI-assisted research drafts. Some claims and prerequisites have checked Lean proofs, mapped precisely in `FORMALIZATION.md`; no complete verification of the collection is claimed. This catalogue describes the reports rather than certifying them. The cross-report `NOTATION.md` explains shared names and distinctions between local conventions.

Rebuild this document with

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latexmk -pdf manifest.tex && latexmk -c manifest.tex
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# 1 The surreal field

Eighteen reports about **No** and its real Hahn subfields. Two questions of Lipparini on evaluating formal sums; two independent halves of Gonshor’s product-birthday conjecture, on domains neither of which contains the other; the subgraph question from Roughan’s paper on birthday arithmetic; a genetic counterexample separating pointwise derivatives from differences of constants; a rigidity theorem for exponential automorphisms; a sharp convexity classification of surreal Gamma functions; differential transcendence of explicit surreals; Hahn-valued measures and probability; Markov generators at every valuation scale; transcendence over bounded-support series; diagonal matrix scaling; vector and tensor fields in every dimension; Euclidean three-space and  $\mathrm{SO}(3, \mathbf{No})$ ; and finite surreal probability; and two new placements on omnific Diophantine geometry and set-sized quotients. Among the first sixteen, the first and the fourth each arrived as a pair of archives and were merged; the exponential-automorphism report later absorbed two independent re-derivations of its own theorem, and the measures report merges five manuscripts, the vector-field report three and the three-space report three; the others are single manuscripts. The Diophantine report assembles three manuscripts, with two further companions placed but unintegrated; the quotient report remains a base text awaiting twelve companions. A further entire-function rectification manuscript is also placed but not yet integrated.

## 1. A square-root obstruction to evaluating increasing Hahn series at $\omega$

[surreal/hahn-evaluation-at-omega/article.pdf](#)

from [hahn\\_evaluation\\_research.zip](#) + [reversed\\_hahn\\_series.zip](#)

Answers Problem 7.7 (p. 38) of Lipparini (arXiv:2505.00424v2) negatively: there is no ring homomorphism from the field of set-supported real-coefficient Hahn series with strictly increasing surreal exponents into the surreals sending  $x^a$  to  $\omega^a$ . The certificate is  $B \in \mathbb{Q}[[x]]$  with  $B^2 = 1 - x$ , built by the recursion  $b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k}$ : any such map would force a square to equal  $1 - \omega < 0$ . The homomorphism is applied only to the finite identity, so no continuity, valuation, order preservation or preservation of infinite sums is assumed. Adds the exact classification of admissible images of the variable as the infinitesimals, the monomial classification over an arbitrary ordered abelian group, and order-uniqueness on the source field. *Merged; the core certificate was identical in both originals and is proved once, but the two toolkits are kept — in particular a direct nonnegative-coefficient semiring certificate. Restriction and extension by differences relate the ring and semiring maps; the two witnesses give distinct elementary proofs.*

## 2. Gonshor’s Product-Birthday Conjecture for ordinal-support series

[surreal/gonshor-product-birthdays/surreal\\_product\\_birthdays.pdf](#)

from [gonshor\\_product\\_birthdays\\_research.zip](#)

Proves Gonshor’s bound  $b(xy) \leq b(x) \otimes b(y)$  on the ring of Conway normal forms  $\sum_{\alpha \in S} c_\alpha \omega^{-\alpha}$  with  $S$  an arbitrary set of ordinals, so transfinite and uncountable supports are allowed. Supplies an exact birthday formula by transfinite induction over the support with maintained prefix bounds and an explicit limit stage, and a complete classification of the equality cases. The route is a leading-Cantor-degree gap, which also gives that the degree of the birthday never rises under multiplication. The conjecture for arbitrary surreal numbers is not solved. The domain is incomparable with that of the Laurent-series report [surreal/gonshor-laurent-birthdays](#): it excludes  $\omega$  and every positive power.

## 3. A Laurent-series case of Gonshor’s birthday conjecture

[surreal/gonshor-laurent-birthdays/article.pdf](#)

from [Gonshor\\_Laurent\\_Birthday\\_Research.zip](#)

Proves the same bound  $b(xy) \leq b(x) \otimes b(y)$  on the canonically embedded field  $\mathbb{R}((\omega^{-1}))$ , which unlike the ring of [surreal/gonshor-product-birthdays](#) contains  $\omega$  and its positive powers but restricts supports to order type at most  $\omega$  on the integer lattice. The route is a bilinear budget: an exact splitting  $b(x) = b(P) \oplus b(u)$  into polynomial and infinitesimal parts, three separate product estimates for the resulting cross terms, and a finite-jet stabilisation carrying the finite bound to

infinite series without any continuity claim. Also treats finite normal forms with arbitrary ordinal exponents, derives exact birthdays for finite and infinite Laurent series, and proves a reciprocal corollary. Neither this theorem nor that one contains the other; on their common corner  $\mathbb{R}[[\omega^{-1}]]$  the two agree and are proved twice.

#### 4. Canonical forms need not be subgraphs

[surreal/canonical-forms-need-not-be-subgraphs/surreal\\_graphs.pdf](#)

from [surreal\\_graphs\\_research.zip](#) + [surreal\\_sparse\\_graphs.zip](#)

Answers negatively the question in the Figure 3 caption of Roughan, *Surreal Birthdays and Their Arithmetic* (arXiv:1810.10373v2, p. 7): the canonical form of a surreal value need not be a subgraph of every other form of that value. Five vertices and rank three are both least possible, and there are exactly sixteen five-vertex counterexamples, eight triangle-free, four of value  $1/2$ . Adds cycle representations of  $1/2$  at every admissible length, the exact extremal function, a planar construction of maximum degree three and arbitrarily large girth, classifications of which canonical graphs are unavoidable, and transfinite witnesses. *Merged; the two originals exhibited genuinely different five-vertex witnesses — one passing through a vertex of value  $-1$ , one with no negative value at all — and both are printed in full, as are two odd-cycle families, two high-girth constructions and two proofs of the Fibonacci bound.*

#### 5. A countable genetic cut across the finite–infinite gap

[surreal/genetic-gaps-and-primitives/article.pdf](#)

from [genetic\\_gap\\_counterexamples.zip](#)

With  $q(t) = t/(1+t^2)$  and  $B(x) = 1 - \{q(n-x) : n \in \mathbb{N} \mid \}$  as a Conway cut, proves that  $B$  is the indicator of the positive infinite surreals, is genetic under the set-sized option constructors of the cited sources, and has pointwise derivative zero everywhere. Consequently  $x$  and  $x - B(x)$  are genetic primitives of 1 agreeing at every real input that do not differ by a constant, the latter also an increasing bijection. The sublevel class  $\{x : 0 < x < \omega, B(x) \leq \frac{1}{2}\}$  is nonempty and bounded yet has no surreal supremum, so neither end gap supplies one.

#### 6. A Number-Valued Broadcast Sum of Surreal Sequences

[surreal/broadcast-sum-of-surreal-sequences/article.pdf](#)

from [surreal\\_broadcast\\_sum.zip](#)

Analyses the sign-truncation game of Lipparini (arXiv:2505.00424v2, Remark 7.6(3), in connection with Problem 7.5). Proves the game always has a surreal-number value, that it agrees with finite surreal sums and with Lipparini’s ordinal operation on their respective domains, and that it fails weak coordinatewise monotonicity, with an explicit family  $s[n] = 3 \cdot 2^{-n-d-1}$  on a subset  $E$  and  $2^{-n-d}$  off it realising the failure. The underlying rule is Lipparini’s; the name and the notation are local to the report.

#### 7. Cofinal displacement and the rigidity of exponential automorphisms

[surreal/exponential-automorphism-rigidity/article.pdf](#)

from three manuscripts, delivered 21 and 22 September 2026

Let  $F$  be an ordered field with a total increasing exponential  $E : (F, +) \rightarrow (F_{>0}, \cdot)$  that is *onto*, and  $w$  a nontrivial convex valuation — no Hahn workspace, completeness, saturation or set-size hypothesis. The engine is one finite lemma: the displacement  $D(x) = \sigma(x) - x$  of a unital field endomorphism is *not* a derivation but obeys  $D(ab) = \sigma(a)D(b) + bD(a)$ , and from  $D(a) = d \neq 0$  the two probes  $b = 2B(1 + |\sigma(a)|)/|d|$  and  $ab$  force  $\max\{|D(b)|, |D(ab)|\} > B$ , with neither surjectivity nor order preservation of  $\sigma$ . Through the surjective, additive, order-reversing map  $\nu = w \circ E$ , a nonidentity exponential automorphism stabilizing  $\mathcal{O}_w$  displaces value-group elements cofinally and coinicially, so  $\rho_w : \text{Aut}_{\text{exp}}(F; \mathcal{O}_w) \rightarrow \text{Aut}(\Gamma, +, <)$  is injective; coarsening rigidity takes stabilization of  $\Delta$  as an explicit hypothesis. The kernel of  $\nu$  is a proper convex subgroup, bounded by a *field* element (the finite surreals by  $\omega$ ), so on **No**, with  $B = \omega$ , an exponential automorphism fixing every natural-valuation value is the identity (Theorem 5.1; the hypothesis is not to be weakened to preserving comparisons), hence so is every exponential 1-automorphism, with no strong-additivity

or coefficient-fixing hypothesis, offered as a negative answer to Question 5.4 of Kaplan–Krapp–Serra, arXiv:2509.22374v3. *Keep that version pinning prominent:* the reading was checked against that PDF on 22 September 2026, which verifies the interpretation, not priority, and Gonshor’s exponential and the natural valuation are cited, not reproved. Section 8 gives two disjoint families of value-group automorphisms, bounded-layer reweightings and positive rational dilations, that lift as ordered field automorphisms but not exponentially, so cofinal displacement is necessary and not sufficient; Section 9, for real closed  $F$ , gives  $\text{Aut}_L(F(i)) \cong \text{Aut}_{\text{exp}}(F) \times C_2$  for  $L(z) = \log \sqrt{z\bar{z}}$ , with no complex-logarithm branch. Two later manuscripts re-derived the engine and the answer by the same proof, printed once; merging them added the theorem that no nonzero twisted derivation of a nontrivially valued field has all output valuations bounded below by one element, at any rank and with no ordering (Theorem 3.5), rigidity from one visible exponential scale without surjectivity of  $E$  (Theorem 4.4), an order-free detector (Corollary 4.5), profile rigidity on a final interval (Theorem 6.1), cofinal multiplicative motion and cofinal commutation defects of value-fixing maps, the latter not asserting that such a map exists (Section 7), no global bound  $v(\partial y) \geq v(y) + \gamma$  for a nonzero  $E$ -compatible derivation of a nontrivially valued field, applied to the Berarducci–Mantova derivation (Theorem 10.3), and  $\mathbb{R}((t))$  sharpness examples that are not counterexamples (Section 11). Rigidity here means faithfulness, not triviality; the image of  $\rho_v$  on  $\mathbf{No}$  is not classified (Question 13.1); the derivation theorems exclude only global bounds. The Berarducci–Mantova derivation is an object, not a theorem, shared with **surreal/gamma-functions** and **surreal/tail-spans-and-differential-transcendence**, and this third “rigidity” is neither the all-scale polynomial rigidity of **surcomplex/analysis** nor the scalar rigidity of **surreal/gamma-functions**. The staged source audit’s conclusion that the collection lacked this report is false, as its appended correction says. Generic displacement, exponential-profile, logarithmic-modulus and valuation-kernel proofs now have Lean mappings; actual surreal exponential instantiations remain pending. Not refereed; no exhaustive priority claim.

## 8. Gamma functions over the surreals: a sharp convexity classification

[surreal/gamma-functions/article.pdf](#)

from one manuscript, delivered 21 September 2026

Write  $M(x)$  for the leading Conway monomial of a positive infinite surreal with its real coefficient stripped and  $\text{pp}(x)$  for the purely infinite part; for an arbitrary class assignment  $a$  on positive infinite monomials, no regularity assumed, set  $h_a(x) = a(M(x)) \text{pp}(x)$  and  $L_a = L_0 + h_a$  over the Taylor–Stirling log-Gamma baseline. The main theorem is a two-sided sharp threshold:  $L_a$  is convex on all of  $\mathbf{No}_{>0}$  iff  $ma(m)$  is finite for every positive infinite monomial  $m$ , and in that case it is *strictly* convex — pointwise in  $m$ , no uniform real bound across monomials, both signs allowed, and no merely-convex member exists. Every member of the family, convex or not, preserves  $\Gamma_a(1) = 1$ , the recurrence  $\Gamma_a(x+1) = x\Gamma_a(x)$ , every finite Gauss multiplication formula, every normalized finite-translation germ and all positive-order fine derivatives of log-Gamma, so the literal surreal Bohr–Møllerup conditions do not give uniqueness:  $a_\lambda(m) = \lambda e^{-m}$ ,  $\lambda \in \mathbb{R}$ , gives pairwise distinct strictly log-convex extensions whose logarithmic and relative differences are smaller than *every* ordinary inverse power at *every* infinite argument, the log-difference at  $\omega$  being exactly  $\lambda \omega e^{-\omega}$ , and adding all ordinary-order Stirling envelopes does not restore uniqueness. Absolute Gamma differences need not be small: at  $\omega$  the difference  $\Gamma_1 - \Gamma_0$  exceeds every ordinary power of  $\omega$ . The convexity proof uses no second derivatives and no mean value theorem: it works with the tangent gap  $D_F(y, x) = F(y) - F(x) - F'(x)(y - x)$  through four exhaustive relative-scale regimes, the margin  $(y - x)^2/(2x)$  being exactly what generates the  $O(m^{-1})$  threshold, and the mechanism is then isolated as a reusable scale-margin gauge principle. A positive second derivative does *not* imply convexity in this calculus — the gauge with value one on one monomial and zero elsewhere has all fine derivatives equal to the baseline’s and a negative tangent gap — because such gauges are finely locally constant and convexity cannot be decided locally. Compatibility with the Berarducci–Mantova derivation forces  $h_a$  into  $\ker \partial = \mathbb{R}$  and kills every member but  $a = 0$ , yet a different real-valued non-flat gauge survives, so *scalar compatibility alone is not a uniqueness characterization*. Closes with an exact finite-phase domain on the tube  $\text{Re}(z) > 0$ ,

$\text{Im}(z)$  finite, using the finite-angle exponential of the imaginary log-Gamma value, distinct from choosing a primitive in [surcomplex/differential-equations](#). *Not offered as a resolution of a named published problem*, and the classification is exact only for the named monomialwise family. Its own audit read an earlier state of this catalogue; its eighteen-package count is a historical description, not the current inventory.

## 9. Tail-spans and differential transcendence in surreal and surcomplex Hahn fields

[surreal/tail-spans-and-differential-transcendence/article.pdf](#)

from one manuscript, delivered 22 September 2026

Let  $M$  have characteristic zero,  $\Gamma$  be a set-sized ordered abelian group,  $(a_i) \subset M^\times$  have independent square classes with  $r_i^2 = a_i$ , the exponents  $\gamma_i$  be distinct with well-ordered image, and  $v_i \in M^r$ ; put  $Y = \sum_i t^{\gamma_i} r_i v_i$  over  $B = M((t^\Gamma))$ . The invariant is the *cofinite span*  $W$ , the intersection of the spans surviving every *finite* deletion, not deletion of a transfinite initial segment. Theorem 5.1:  $E = \{i : v_i \notin W\}$  is finite and  $B[Y_1, \dots, Y_r] = B_E[s_1, \dots, s_d]$ , where  $d = \dim_M W$ ,  $B_E = M(r_i : i \in E)((t^\Gamma))$  has degree  $2^{|E|}$  over  $B$  and the  $s_j$  are algebraically independent over  $B_E$  — an equality of subrings, not of dimensions; Theorem 5.2 splits the relation ideal of  $Y$  over  $B$ , after base change to  $B_E$ , into  $2^{|E|}$  pairwise comaximal affine planes of dimension  $d$ . The proof is finitary: the alternating sum of finitely many coefficient sign changes of a degree- $D$  relation is a mixed difference equal to  $D!$  times its top polarization, which cofinite spanning forces to vanish identically. With  $t = \omega^{-1}$ ,  $B_0 = \mathbb{Q}((t^\mathbb{Q})) \subset H_0 = \mathbb{A}_\mathbb{R}((t^\mathbb{Q})) \subset \mathbf{No}$  (real-algebraic coefficients) and  $\xi_A = \sum_{n \in A} \sqrt{p_n} \omega^{-n}$ ,  $p_n$  the  $n$ th prime: for *every* infinite  $A$  the Berarducci–Mantova jets  $\partial^k \xi_A$  are algebraically independent over  $B_0$  (Theorem 7.1), and those of the explicit binary-prefix almost disjoint family  $(\xi_\alpha)_{\alpha \in \{0,1\}^\mathbb{N}}$  jointly (Theorem 7.3); with a separate count  $|H_0| \leq 2^{\aleph_0}$  for the upper bound,  $\text{dtrdeg}_{B_0} H_0 = \text{trdeg}_{B_0} H_0 = 2^{\aleph_0}$ . For every infinite  $A$ , all mixed jets  $\partial_z^j \delta^k F_A$ ,  $\delta = t d/dt$ , of  $F_A = \sum_{n \in A} t^n \sqrt{1 - z/2^n}$  are independent over  $\mathbb{C}(z)((t^\mathbb{Q}))$  (Theorem 9.3), and continuum many such functions jointly on the one disk  $|z| < 1$  (Theorem 10.1). Also a value-group-relative valuation-closure criterion (Theorem 12.1) and undecidability of algebraicity and differential algebraicity over  $B_0$  for total computable coefficient streams (Proposition 13.1). Cautions: the sign changes are not order-preserving and are not extended to  $\mathbf{No}$ ;  $\mathbb{C}(z)((t^\mathbb{Q}))$  is not a common-domain ring; point values are not claimed transcendental;  $F = F_{\mathbb{N}_{>0}}$  satisfies  $F(2z, t) - tF(z, t) = t\sqrt{1 - z}$ , so functional equations are not excluded; the family indexed by *all* subsets is dependent,  $\xi_{A \cup B} + \xi_{A \cap B} = \xi_A + \xi_B$  (Remark 7.4, added on fitting). The formula for  $\partial$  on  $\mathbb{C}((t^\mathbb{Q}))$  is cited from [surcomplex/differential-equations](#); only the restriction to real-algebraic coefficients is checked here. [surcomplex/dynamics-and-normal-forms](#) keeps, unweakened, its warning that its  $H_\varepsilon$ , with  $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$ , is not differentially transcendental; Theorem 9.3 concerns different functions. Its 404 exact finite checks prove no infinite statement; priority not certified, not refereed. The Lean ledger records proofs of cofinite-span stabilization and multilinear detection; the classification and differential-independence theorems remain pending.

## 10. Hahn-valued measures and probability: strong atomicity, coefficientwise extension, and hidden negative mass

[surreal/hahn-valued-measures-and-probability/article.pdf](#)

from five manuscripts, delivered 22 September 2026

Measures with values in  $\mathbb{R}((t^\Gamma))$  and  $\mathbb{C}((t^\Gamma))$ , and through  $t^\gamma = \omega^{-\gamma}$  in  $\mathbf{No}$  and  $\mathbf{No}[i]$ , with  $\Gamma$  a set-sized ordered abelian group *not* assumed divisible except in one Gaussian quadrature theorem, under two meanings of countable additivity: a *coefficientwise* Hahn measure is a Hahn series of ordinary finite signed or complex measures on one well-ordered exponent set, added exponent by exponent; a *strong* one requires the masses of every disjoint sequence of events to be strongly Hahn summable. **Strong class.** On every countably separated space a strong measure is  $\mu(A) = \sum_{x \in A}^H w_x$  for one strongly summable family  $(w_x)$ , common support and atomicity being conclusions; it extends uniquely to all subsets, and a strong probability has finitely supported standard part. Exact criteria are proved for coherent cylinder data over finite alphabets, for independent finite-state coordinates (finitely many random at the residue level, and the individual



rare entries, not their aggregates, strongly summable), and for moments on an ordinary real sample set. One weight pattern of support order type  $\omega + 1$  gives nonnegative Cantor-space cylinder masses, or a functional strictly positive on nonzero Hahn polynomials nonnegative on  $[0, 1]$ , yet a negative singleton;  $\omega + 1$  is least. **Coefficientwise class.** A transfinite null-ideal criterion decides positivity; two exponents already hide a negative singleton; independent Bernoulli perturbations on a common well-ordered exponent set extend exactly under weighted  $\ell^2$  rows at baselines uniformly separated from 0 and 1, and  $\ell^1$  rows at deterministic ones. *The threshold  $\omega + 1$  and the finitely supported standard part hold in the strong class only.* Sources 10 and 11 proved the classification independently, and the threshold for cylinder and polynomial tests respectively; each is printed once, 10 as base; source 15's classification on compact metric spaces, with a common support *assumed*, is recorded as a special case. Not a first non-Archimedean Kolmogorov theorem (Ludkovsky–Khrennikov proved one under other axioms); priority not certified and not refereed. The Lean ledger records checked strong atomicity and product prerequisites; the complete report is not formalized. **physics/surreal-scalars-and-spacetime** asked for a summation theory, and this report works out two; it shows that the object refused by non-claim N75 of **surcomplex/dynamics-and-normal-forms** cannot exist in the strong class for an ordinary group action with only infinite orbits, and states canonically the null-ideal criterion proved independently in **surcomplex/hahn-herglotz-positivity**. A fourth manuscript (Sections 25–31) develops the coefficientwise class for countably many scales: every measure has a Hahn-valued density with respect to one finite real control measure, positivity is positivity of that density almost everywhere (related exactly to the null-ideal criterion, whose proof it re-derives for countable exponent sets), and Hahn–Jordan, Radon–Nikodym, Fubini, conditional expectation and infinite masses follow. A fifth manuscript (Sections 32–34) removes the uniform separation of the Bernoulli product theorem: at arbitrary interior baselines the laws extend exactly when every exponent row lies in  $\ell^1 + \ell^2(1/[p_j(1 - p_j)])$ , answering the report's near-boundary question; a mixed-baseline version is added by the merge and marked as such.

## 11. Surreal Markov generators at every valuation scale

[surreal/markov-generators-at-every-scale/article.pdf](#)

from one manuscript, delivered 22 September 2026

Finite Markov chains with positive Hahn rates  $q_e \in F_{>0}$ ,  $F = \mathbb{R}((t^\Gamma))$ , on a strongly connected digraph with  $n \geq 2$  states;  $\Gamma$  is nonzero, set-sized and *divisible* throughout, read in **No** through  $t^\gamma \mapsto \omega^{-\gamma}$ . The object is the normalized resolvent  $R_L(s) = s(sI + L)^{-1}$  of the row Laplacian  $L$  (generator  $-L$ ), defined rational-algebraically, and its residue matrix  $K_\alpha(c) = \text{res } R_L(ct^\alpha)$  at each scale  $\alpha \in \Gamma$ , the report's *shadow*, unrelated to the shadow topology. In-forests give the valuation and leading coefficient of *every* resolvent entry, zero-residue entries included (Theorem 4.1). There are at most  $n - 1$  crossover scales, and the plateaux form a flag of distinct stochastic idempotents  $I = P_0, \dots, P_m = \mathbf{1}p_0$ ,  $P_i P_j = P_j P_i = P_{\max(i,j)}$ , of strictly decreasing rank (Theorem 5.2). On the canonical stochastic retract  $P = UV$ ,  $VU = I$ , each crossover is  $U c(cI + G_\alpha)^{-1} V$  for a real Markov row Laplacian  $G_\alpha = (V K_\alpha(1) U)^{-1} - I$  with semisimple zero eigenvalue, with no reversibility, simple-eigenvalue or unique-minimizing-forest assumption (Theorem 6.2). Conversely (Theorem 7.4, Corollary 7.5), every such flag ending in rank one, with any compatible real effective generators, occurs at any prescribed scales  $\alpha_1 < \dots < \alpha_m$  in an irreducible Hahn row Laplacian, whose intermediate-scale positivity corrections add no crossover. Also: relative rate errors of valuation  $\geq \delta$  on one fixed graph give the same bound for every resolvent entry, uniformly in  $s > 0$  (Theorem 8.1; sharp by Example 8.3); a real one-parameter monomial model reproduces the whole *finite* marked-forest leading diagram, so such data cannot certify higher rank (Theorem 9.2); and  $\det(cI_p + G_\alpha) = cD_\alpha(c)/b_{k_-}$  yields the leading surcomplex eigenvalue amplitudes of  $L$ , including tied scales and nonreal eigenvalues, while the forest profile is convex (Theorems 10.2–10.3). *One manuscript; no merge:* placement collected its own limitations as (N1)–(N11) and added a divisibility ledger and cross-references, changing no theorem number. Not treated: a global surreal exponential, path measures, infinite state spaces, nondivisible  $\Gamma$ . Classical forest, stochastic-idempotent and rank-one metastability inputs are credited; priority is not

certified; not refereed. The Lean ledger records checked algebraic clauses of the resolvent identity, projection-flag construction and irreducible completion; residues and the remaining hierarchy theorems are pending. The finite exact checks are not proofs. Lemma 10.1 is the forest-polynomial case of the Newton rule of `surcomplex/polynomial-algebra`; the positivity mechanism parallels the Cauchy–Binet identity of `surcomplex/spectral-theory` for another matrix class; neither this report nor `surreal/hahn-valued-measures-and-probability` depends on the other.

## 12. Transcendence over bounded-support Hahn series

[surreal/transcendence-over-bounded-support/article.pdf](#)

from one manuscript, delivered 22 September 2026

Let  $G$  be a nonzero set-sized ordered abelian group,  $K$  a field of characteristic zero,  $\mathcal{B}_G(K)$  the ring of Hahn series whose support is bounded above,  $\mathcal{F}_G(K)$  its fraction field and  $\kappa = \text{cf}(G)$ , cofinality in exactly the sense of `surcomplex/entire-functions-at-arbitrary-rank`. There are  $2^\kappa$  elements of  $K((t^G))$ , with positive integer coefficients and one common support of order type  $\kappa$ , algebraically independent over  $\mathcal{F}_G(K)$  (Theorem 5.3); no smaller support cardinality carries even one transcendental element (Corollary 5.4), every valuation ball contains such a family, and for real-exponent Hahn fields over  $\mathbb{R}$  or  $\mathbb{C}$  the transcendence degree is the continuum. A descent theorem makes  $K((t^H))$  and  $\mathcal{F}_G(L)$  linearly disjoint over  $\mathcal{F}_H(K)$  for cofinal  $H \subseteq G$  and  $K \subseteq L$  (Theorem 3.2), so relation ideals and finite degrees survive enlargement of coefficients and cofinal exponents. Also: when  $\mathcal{B}_G(K)$  is a field, its units via the highest Archimedean quotient, a GCD-descent theorem that stays *conditional*, and Conway-normal-form witnesses that are actual surreal numbers. Transcendence here comes from the support, not from the coefficients as in `surreal/tail-spans-and-differential-transcendence`; the two bases do not contain each other, and neither report uses the other. One manuscript; priority not certified, not refereed, no Lean verification.

## 13. Diagonal matrix scaling at surreal scales

[surreal/matrix-scaling-at-surreal-scales/article.pdf](#)

from one manuscript, delivered 22 September 2026

Diagonal scaling of a positive matrix to fixed row and column sums over  $F = \mathbb{R}((t^\Gamma))$  and  $K = \mathbb{C}((t^\Gamma))$ , whose entries may sit at arbitrarily different infinitesimal scales; the local theory assumes no rank or divisibility for  $\Gamma$ . The normalization has a unique support-controlled analytic branch on the whole relative infinitesimal polydisc. The invariant is the *spanning-tree deletion gap*: with  $\tau$  the least valuation sum over spanning trees of the bipartite support graph and  $\tau_e^-$  the least over trees avoiding the edge  $e$ ,  $\kappa_e = \tau_e^- - \tau$  is the exact uniform gain at entry  $e$  — an input change of valuation at least  $\delta > 0$  moves that output entry by valuation at least  $\delta + \kappa_e$ , a nonlinear two-point estimate, attained by changes in  $e$  alone, with bridges fixed exactly. Taylor coefficients obey the same bounds; evaluation at arbitrary rank needs a separate finite-generator support argument; along a tridiagonal bistochastic chain the gains add exactly. The positive-sum mechanism is that of the forest sums of `surreal/markov-generators-at-every-scale` and the Cauchy–Binet sums of `surcomplex/spectral-theory`, but the maps differ: the Markov report’s stability bound has no gain, and  $\kappa_e$  is not a condition number. No convergence or complexity claim for Sinkhorn iteration. One manuscript; not refereed, no Lean verification, priority not certified.

## 14. Surreal vector and tensor fields in every dimension

[surreal/vector-and-tensor-fields/article.pdf](#)

from three manuscripts, delivered 22 September 2026

Replacing  $\mathbb{R}$  by **No** gives tensor algebra but no calculus. Three manuscripts built the same layered theory, printed once from the  $n$ -dimensional one: finite algebra over a set-sized real-closed surreal subfield; a sheaf of locally Hahn-supported smooth tensor fields on ordinary real spaces, under one support condition that makes sums, products, contraction, differentiation and integration over compact real chains compatible; and genuine surreal-coordinate neighbourhoods reached by support-controlled Taylor prolongation and scale charts. Part I gives Lie and Cartan calculus,

connections and curvature, Hodge theory for a real or regular Riemannian Hahn metric, Stokes and Poincaré, Hahn cohomology, standard-part reduction and an infinite-frequency obstruction. Part II, from the three-dimensional manuscript, is dot and cross products, integral theorems and multiscale continuum equations; Part III, from the Minkowski manuscript, is Lorentz transformations at every scale, the Lorentzian Hodge operator, Maxwell and scalar fields and stress–energy, as mathematics on Minkowski space — not an assessment in the sense of **physics/surreal-scalars-and-spacetime**. Results proved in three dimensions are printed in every dimension only where the proof allows, and nothing Lorentzian is attached to a general statement. The complex-coefficient Stokes and cohomology results of **surcomplex/contours-and-stokes** are credited, not reclaimed. Not refereed, no Lean verification, priority not certified.

## 15. Euclidean three-space and the rotation group over the surreals

[surreal/euclidean-three-space/article.pdf](#)

from three manuscripts, delivered 22 September 2026

Coordinate geometry of  $\mathbf{No}^3$  inside set-sized real-closed Hahn fields, where lengths, areas and radii may be infinite or infinitesimal but angles between directions always have canonical finite representatives, so spatial trigonometry needs no oscillatory meaning for an infinite angle. Part I, the base: incidence, projections, skew-line distance, sphere intersections, quadrics, rigid motions, Rodrigues rotations, screw motions, spherical coordinates, and spherical trigonometry with sine and cosine laws, branch-correct excess, area and holonomy. Part II is the class group  $\mathrm{SO}(3, \mathbf{No})$ : every nonidentity rotation has an axis, Rodrigues, Euler and Cayley coordinates stay exact, the standard part gives a split extension  $K \rtimes \mathrm{SO}(3, \mathbb{R})$  whose infinitesimal kernel is uniquely divisible and perfect (every element a single commutator), finite subgroups are conjugate to real ones, and the rotation exponential has rank-one critical spheres. What both manuscripts proved (isometries, Rodrigues, the Cayley chart, the finite phase) is printed once; the spin double cover is cited from **surquaternions/surquaternions**, and the plane case is the  $\mathrm{SO}(2, \mathbf{No})$  section of **surcomplex/trigonometry**. Not refereed, no Lean verification, priority not certified. A third manuscript (Part III) classifies the normal subgroups of  $\mathrm{SO}(3, F)$  by valuation cuts, computes lower central and derived series, and proves that every class homomorphism from  $\mathrm{SO}(3, \mathbf{No})$  to a set-sized group factors through standard part, also for  $\mathrm{SO}(n, \mathbf{No})$  and  $\mathrm{SU}(n, \mathbf{No}[i])$ .

## 16. Finite surreal probability, log-odds and conditioning

[surreal/finite-surreal-probability/article.pdf](#)

from one manuscript, delivered 22 September 2026

Probability with values in set-sized subfields of  $\mathbf{No}$ , with logarithmic odds from the surreal exponential. On finite event algebras, where finite, strong and coefficientwise additivity coincide: conditioning on infinitesimal events and exact Bayesian updates, a leading-scale calculus with coefficient-sensitive tie breaking, quantitative conditioning stability, softmax models, relative entropy, proper scoring rules, Gibbs variational principles and finite-horizon processes; a conditional real shadow keeps what the unconditional standard part destroys. Beyond finite algebras it gives regular finitely additive laws on all subsets for two coin models, a fine-ultrafilter law, a Loeb bridge, and a counterexample to positive coefficientwise integration of Hahn-valued functions. Its coin theorems are the cases  $p = t$  and  $p = \frac{1}{2} + t$  of the i.i.d. corollary of **surreal/hahn-valued-measures-and-probability**, which it answers no question of. No physical interpretation is claimed, in line with **physics/surreal-scalars-and-spacetime**. One manuscript; not refereed, no Lean verification, priority not certified.



## 17. Omnific integers and omnific–Diophantine geometry

[surreal/omnific-diophantine-geometry/article.pdf](#)

from 01, 02 and 05 assembled in bdd536; 06–07 placed in cf350b1

Arithmetic and Diophantine geometry over Conway’s omnific integers: constant-term retraction, equational transfer, finite quotients, linear systems, binary and norm-form rigidity, quadratic levels and Diophantine definitions of the ordinary integers. Commit bdd536 assembles sources 01, 02 and 05, including Euler derivations, separated powers, unimodular Fermat rigidity and two quartic definitions. The proof review reaches Section 10, including support and class-quotient clarifications, complex separated-power rigidity, quadratic coordinate-ideal preservation, quantified bounded-set transfer, and explicit induction domains and workspace arguments. Later proofs and remaining source reconciliation await review. Two further definability and reconstruction manuscripts remain unintegrated. A current PDF and maintained build instructions accompany the source. No new Lean coverage is claimed.

## 18. The universal set-sized quotient of the omnific integers

[surreal/set-sized-quotients-of-omnific-integers/article.tex](#)

from 03, 04 and 06–11 placed in be06fc8; 12–16 in cf350b1

The base manuscript claims that every unital ring map from the omnific integers to a set-sized ring factors through the constant term; it also studies Gaussian variants and cardinal and homological thresholds for set-sized integer parts. Only base manuscript 06 is currently installed as the article. Twelve companion manuscripts, delivery filenames and current PDF build await integration; proof review and formalization remain pending. The catalogue does not certify the delivery’s priority claims.

## 2 The surcomplex plane

Twenty-six reports about  $\mathbf{No}[i] = \mathbf{No}[i]$ . Seven are about holomorphic function theory, rest on one framework, and are all merges: twenty-seven manuscripts reduced to seven documents, with three more later added to trigonometry, one on  $\mathrm{SO}(2, \mathbf{No})$  and two on the periods of global phases. The other nineteen came later and are listed after them. Nine are merges: differential algebra (eleven manuscripts), dynamics and normal forms (ten), nonabelian support obstructions (three), entire functions at arbitrary rank (six), finite spectral theory (three), Hahn–Tate uniformization (three), holonomic rigidity (five), infinite-dimensional spectral theory (four) and Gamma and zeta (six). Ten are single manuscripts: on rank-one analytic geometry, expanding polynomial dynamics, Herglotz positivity, moment reconstruction, Wick summability, the three duals of Hahn vector spaces, hidden negative Hermitian directions, the automorphisms of the field, cardinally bounded Hahn fields, and what one dilation defines. Each of the nineteen was written against this repository and names the gap it fills.

*Read **analysis** first.* It is the foundation the other six continue. In the *full fine topology*, every set-sized subset is closed and discrete and every convergent set-indexed net is eventually constant. Universe-relative formulations require smaller-universe subsets and indices. These conclusions do not apply to the intrinsic valuation topology of a fixed Hahn field:  $t^n \rightarrow 0$  in  $\mathbb{C}((t^{\mathbb{Q}}))$ . Strong Hahn summability is a separate support condition; even in rank one it need not give convergence of finite subsums, as the foundations report’s  $\sum t^{n/(n+1)}$  demonstrates. Contour integration in the analytic reports is defined coefficientwise on ordinary complex paths.

*Then read the ring table in **analytic-geometry**, before anything else.* Four coefficient rings circulate through this material under nearly identical notation — the radius-free  $\mathbb{C}\{z\}((t^{\Gamma}))$ , the fixed-polydisk  $\mathcal{O}(D)((t^{\Gamma}))$ , the common-domain germ ring, and the formal  $\mathbb{C}[[z]]((t^{\Gamma}))$ . These are not an interchangeable chain. For  $\eta > 0$ , the family  $F(z) = \sum_{r \geq 1} t^{r\eta}/(1 - rz)$  lies in the radius-free germ ring but not the common-domain germ ring: its coefficient poles  $1/r$  approach zero. This witnesses that particular strict inclusion. The report gives separate fixed-domain obstructions and explains which local proofs require shrinking. The remaining five reports can be read in any order.

## 19. Surcomplex analysis: holomorphic functions on $\mathbf{No}[i]$

[surcomplex/analysis/article.pdf](#)

from `surcomplex_analysis.zip` and eight companion archives

One article assembled from nine manuscripts delivered on 21 September 2026. It develops Hahn-summed local germs, coherent sections over connected ordinary complex domains, preparation and finite zero clusters, contour residues, exponentials and logarithms, and polynomial or rational rigidity under coherence at every positive surreal scale. Each formal series has a sufficiently small realization, depending on the series; evaluation is faithful as a function germ, not at one fixed point. The uniform Taylor-remainder bound gives the full fine derivative. Canonical lifts, coherent sections and germ-analytic functions form strictly increasing classes; germ analyticity implies fine differentiability, with no converse or strictness theorem for that last inclusion here. Coherence supplies global identity, maximum and open-mapping principles over the ordinary base. Sharp Schwarz–Pick bounds require canonical lifts. An ordinary-real value bound and coefficientwise boundedness are incomparable:  $tZ$  satisfies the first but not the second, while the constant  $\omega$  satisfies the second but not the first. Only the coefficientwise Liouville hypothesis forces constancy; one arbitrary surreal bound is always available. For  $1/(\zeta - \varepsilon)$  with nonzero infinitesimal  $\varepsilon$ , the cluster residue at the ordinary centre 0 is 1, while the actual local residue at 0 is 0. Rouché’s halo theorem requires an ordinary margin; a bare strict surreal boundary inequality can miss a zero in a boundary monad. The global and cut-plane logarithms differ by  $2\pi i$  on specified points. Every explicit exponential twist retains enough cofinal periods to give the proved value distribution of  $E_\lambda(1/z)$ , although their kernels need not be nested; this is not a general Picard theorem. Full fine topology and the intrinsic topology of a fixed Hahn workspace remain distinct. The nine finite verification programs and historical `MERGE_NOTES.md` are preserved; the maintained `README` identifies corrected comparisons in that older record.

## 20. Infinitesimal analytic geometry over the surcomplex numbers

[surcomplex/analytic-geometry/article.pdf](#)

from `Surcomplex_Analytic_Geometry.zip` + `surcomplex_finite_geometry.zip`

The ring theory of the infinitesimal neighbourhood — the monad  $\mathfrak{m}_K^n$  of an ordinary point — rather than the deformation theory of the reports that follow. Proves, *separately of each of* the radius-free  $A_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$  and the common-domain  $\mathcal{R}_n$ , that it is a Noetherian Jacobson domain of dimension  $n$  whose maximal ideals are exactly the infinitesimal evaluation ideals  $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$  with residue field  $K$ , regular local at each, with completion  $K[[X_1, \dots, X_n]]$  — and that neither is local, each recording every infinitesimal point at once. Carries Weierstrass division and preparation, a weak and a strong Nullstellensatz, Noether normalization, finite projection, contour residue duality, Hensel–Rouché, and workspace invariance: every common zero over any larger divisible workspace already lies in  $K_\Gamma^n$  with unchanged multiplicity, so the count accounts for all zeros in the whole proper class  $\mathbf{No}[i]$ . The nonzero-value-group hypothesis is essential and not aesthetic — for  $\Gamma = \{0\}$  the strong Nullstellensatz fails at  $I = (0)$ . *This is the report that fixes the four-ring vocabulary and proves the inclusions strict; statements holding over both  $A_n$  and  $\mathcal{R}_n$  are stated twice with both proofs, since the two proofs use different resources and neither transfers.*

## 21. Finite Hahn deformations: division, conservation of multiplicity, residue duality

[surcomplex/finite-deformations/article.pdf](#)

from `surcomplex_finite_deformations.zip`, `_finite_fibres`, `_finite_maps`, `_intersections`, `_research`

Five manuscripts on one theorem: an ordinary holomorphic complete intersection  $f$  on a fixed polydisk, with the origin as its only common zero and colength  $\mu$ , keeps its entire finite geometry under any Hahn perturbation  $F = f + E$  of well-ordered *strictly positive* error support — on the unshrunk domain, with an explicit support certificate  $\text{supp} \subseteq \text{supp}(G) + M$ ,  $M$  the monoid generated by the error support. The remainder of division is unique; the witnesses need not be, their ambiguity being a Koszul syzygy carried by the same certificate. Gives the integral structure  $H_\Gamma^+(D)/(F) \cong (K^\circ)^\mu$  over a valuation ring that may have high-rank value group and need not

be Noetherian, with nothing completed; conservation of multiplicity at the *actual* displaced surcomplex zeros rather than their standard parts,  $\sum_{\zeta} \mu_{\zeta} = \mu$ ; and a coefficientwise residue  $\text{Res}_F$  — defined by a series, not by a surcomplex contour integral — whose Gram determinant is a unit even at a collision and which satisfies  $\text{Res}_F(G \cdot J_F) = \text{Tr}(M_G)$ . The separate residue formula, when all zeros are simple, is  $\text{Res}_F(G) = \sum_{\zeta} G(\zeta)/J_F(\zeta)$ ; for nonnegative-support  $G$ , individual terms can be infinite while the total is finite. Also Hartogs, characters and local algebras, spectral invariants, trace-Jacobian and Bézoutian, monodromy and stability. For  $\mu = 1$ , the inverse-map proof constructs one uniform Hahn support before evaluation. The engine throughout is positive-support Neumann inversion; ordinary analytic convergence occurs in finite-parameter comparisons, while the Hahn constructions use strong support summation.

## 22. Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane

[surcomplex/contours-and-stokes/article.pdf](#)

from `surcomplex_contours.zip` and two same-named archives

Three manuscripts whose titles differ by one word — *Stokes Calculus, Integration, Formulae* — over one body of work. All three refuse to transfer the Jordan–Cauchy–Stokes package to  $\mathbf{No}[i]$  as a unit, and all three split it the same way: a coarse Jordan separation theorem in the standard-part topology whose separator is the whole boundary monad rather than a curve; an imported semialgebraic Jordan theorem; a Hahn-coefficient de Rham complex with coefficientwise Stokes, extended to infinitesimally deformed chains; and the holomorphic consequences — Cauchy, endpoint displacement, integer winding, and microscopic circle and torus realizations of the perturbation residue — with an algebraic rational contour pairing characterized uniquely by vanishing on exact differentials and normalizing  $dz/(z - a)$  to  $2\pi i \cdot \text{wind}_{\text{sa}}$ . Adds a purely algebraic third chain category over any characteristic-zero field, a support-preserving Poincaré lemma, an order-valued ML estimate, and a root-of-unity/Schnirelman section. This is where the question the deformation report leaves open is answered: supplying an actual surcomplex contour that represents the residue series. *Two thirds of each manuscript was the same mathematics in the same order; 82 numbered results became 59, by duplicate elimination only.*

## 23. Global support obstructions: moving divisors, Mittag–Leffler, and a Picard dichotomy

[surcomplex/global-divisors/article.pdf](#)

from `surcomplex_global_support`, `_global_theorems`, `_compact_curves`

The local-to-global theory: when do infinitely many individually legitimate infinitesimal prescriptions admit one coherent global realization over the ordinary complex plane? The sharp answer for divisors is a *symmetric-support* criterion — a cluster family  $P_c(u) = u^{m_c} + \sum_{j < m_c} a_{c,j} u^j$  over a closed discrete ordinary  $A \subset \mathbb{C}$  is the exact zero divisor of a nonzero global section iff  $\bigcup_c \bigcup_j \text{supp}(a_{c,j})$  is well ordered: the supports of the symmetric coefficients, not of separately chosen roots. Hence the two discriminating examples —  $n + t^{1/n}$  alone is not realizable, its descending symmetric support  $\{1/n\}$  failing, while *all* roots of  $(z - n)^n = t$  are, symmetric support the singleton  $\{1\}$  — and hence a globally principal effective divisor with a nonprincipal effective sub-divisor. Also Mittag–Leffler at fixed centres and for moving poles, a support-restricted Chinese remainder theory with an explicit Neumann right inverse, an exact Cousin criterion in terms of singular support, and the Picard dichotomy: every ordinary-base stalk is a principal ideal domain, nonlocal when  $\Gamma \neq 0$ ;  $\text{Pic}(\mathbb{C}, \mathcal{H}_{\Gamma}) = 0$  iff  $\Gamma = \{0\}$  or  $\Gamma = \mathbb{Z}\delta$ , and otherwise  $\dim_{\mathbb{C}} \text{Pic} \geq 2^{\aleph_0}$ , with equality for countable noncyclic  $\Gamma$ . A third manuscript studies a compact connected ordinary Riemann surface  $X$  of genus  $g$ . Its Picard group splits into  $H^1(X, \underline{\Gamma}) \oplus \text{Pic}(X) \oplus (H^1(X, \mathcal{O}_X) \otimes_{\mathbb{C}} \mathfrak{m}_{\Gamma})$ . The valuation class is exactly the obstruction to a nonzero coherent meromorphic section, even in arbitrarily high degree. In the zero-valuation sector a finite matrix with positive-support entries computes cohomology and gives Riemann–Roch. A root-free Newton-sum Abel functional decides principality of finite signed clusters. The one-scale comparison has kernel  $H^1(X, \mathcal{O}_X) \otimes_{\mathbb{C}} \bigcap_n t^{n\delta} \mathcal{O}_{\Gamma}$ ; it is injective exactly when  $g = 0$  or  $\delta$  is cofinal in  $\Gamma$ . Any prescribed set of surreal cutoffs can be defeated after enlarging the workspace; all positive cutoffs detect classes in a fixed nonzero

workspace. Coordinate charts transfer the local PID argument to  $X$ , but the global plane and compact classification theorems retain their different base hypotheses.

## 24. Surcomplex polynomial algebra: root geometry, Newton profiles, residue duality

[surcomplex/polynomial-algebra/article.pdf](#)

from `surcomplex_polynomial_algebra.zip` and two same-named archives

Finite-degree polynomial algebra over  $\mathbf{No}[i]$ , worked inside set-sized algebraically closed Hahn workspaces and keeping two geometries deliberately apart throughout: the surreal-valued modulus, whose disks are ordered, and the natural Hahn valuation, whose balls have arbitrary ordered-group rank. A Rouché theorem in each. Covers the finite polynomial algebra and the fundamental theorem there, CRT/Hermite jets, resultants and discriminants, ordered root geometry (root bounds, Gauss–Lucas, Jensen disks, a sharp Schoenberg second-moment inequality with its equality case), real-closed-field transfer with Hermite signature counting, exact root loci under coefficient uncertainty in both geometries, Newton profiles over an arbitrary ordered exponent group, image balls and branch values, exact Rolle counts and the finite root-cluster tree, support-controlled coprime Hensel factorization with a fixed-domain parameter form that shrinks no neighbourhood, and universal residue duality with the trace identity. *Three perturbation theorems with three different hypotheses are kept separate rather than collapsed, as are two discriminant layer formulas and two mapping theorems, with their agreement proved; the alternative was to lose what each source reached that the others did not.*

## 25. Trigonometry on the surcomplex plane

[surcomplex/trigonometry/article.pdf](#)

from `surcomplex_trigonometry.zip` and two same-named archives

Its organizing theorem is that every surcomplex direction has a *finite* surreal angle:  $\text{cis} : (\mathcal{O}_{\mathbb{R}}, +) \rightarrow (T(\mathbf{No}), \cdot)$  is surjective with kernel exactly  $2\pi\mathbb{Z}$ , so  $\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong T(\mathbf{No}) \cong T(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +)$ , and the whole of plane, spherical and hyperbolic triangle geometry is available at arbitrary surreal length scale without ever choosing a value for  $\sin \omega$ . From it: canonical sine and cosine by Hahn summation, the unit circle with its branch conventions and projective half-angle chart, inverse functions on their full surreal domains, the angle-sum, cosine, sine, tangent, half-angle, Heron, incircle, Euler, angle-bisector, cevian, Ceva, inscribed-angle and Ptolemy theorems, an exact valuation dictionary between sides, angles and area with a quadratic triangle-inequality defect formula and two flat-triangle theorems, a rank-two collision-stable intersection algebra, the canonical complex extension to the strip where sine is surjective with completely determined fibres, finite Fourier inversion, Parseval and a surcomplex Fejér–Riesz factorization, and the classification of all global phase extensions, with the Ehrlich–Kaplan normalization placed inside that classification. Two coherence obstructions close it: there is no coherent bounded sine of infinite frequency, and all-scale coherence forces a polynomial. *No mathematical disagreement was found between the three sources; the duplication was near-total on the first two thirds and the divergence near-total on the last.* A later manuscript adds Section 18 on  $\text{SO}(2, \mathbf{No}) \cong \mathcal{O}/2\pi\mathbb{Z} \cong \text{SO}(2, \mathbb{R}) \times (\mathfrak{m}, +)$ : isometries are automatically linear, a rational projective chart, torsion  $\mathbb{Q}/\mathbb{Z}$ , one real coordinate per valuation layer, a strongly summable logarithm, non-algebraic group automorphisms and the fine topology; what it shares with the circle-group results above is printed once. Two further manuscripts (Section 19) study the periods of global phases: their profinite defect, and normalized surcomplex exponentials whose kernel-multiplier ring is exactly  $\mathbb{Z}$  without being an integer part, which refutes one sufficiency possibility in the first Ehrlich–Kaplan robustness question; the question itself stays open.

## 26. Differential algebra and differential equations over the surreal and surcomplex numbers

[surcomplex/differential-equations/article.pdf](#)

from eleven manuscripts, delivered 21 and 22 September 2026

Eleven manuscripts on the Berarducci–Mantova derivation  $\partial$  on  $\mathbf{No}$ , normalized by  $\partial\omega = 1$ , and its unique extension to  $\mathbf{No}[i]$  — unique because differentiating  $i^2 + 1 = 0$  forces  $\partial i = 0$ . Seven share a twelve-item spine, printed once; four later members each add a part. The organizing identity is exact: the logarithmic-derivative image is  $\mathbf{No} + i\mathcal{A}$ ,  $\mathcal{A} = \partial\mathfrak{m} = \partial\mathcal{O}$ , the two forms agreeing only through a proved lemma, so  $\partial y = (a + ib)y$  has a nonzero surcomplex solution exactly when  $b$  has a *finite* primitive.  $\mathcal{A}$  is a *strict* subclass of  $\mathfrak{m}$ : the infinitesimal  $\omega^{-1}$  has the infinite primitive  $\log \omega$ , so  $\partial y = i\omega^{-1}y$  has only  $y = 0$  — collapsing  $\partial\mathcal{O}$  to  $\mathcal{O}$  would make it solvable, which the first eight sources all refute. Hence no formal oscillator:  $\partial y = iy$  and  $\partial^2 y + y = 0$  have only  $y = 0$ . Also the purely infinite part of the accumulated phase as a complete gauge invariant in a four-term exact sequence; the sharp threshold  $\text{minsupp } b > 1$  over real-exponent Hahn fields, with a counterexample pair against globalizing it by valuation; Picard–Vessiot extensions with diagonal phase tori; and a separate coordinate half with a coherent nonlinear initial-value theorem. Part III (member 10): every  $\partial y = Ay$ ,  $A \in \text{Mat}_n(\mathbf{No}[i])$ , is gauge equivalent to  $i \text{diag}(\partial P_j)$ , the multiset  $\text{Ph}(A) = \{P_j\}$  of purely infinite surreals being a complete invariant (Theorem 15.9), computed for Hermitian coefficients from the algebraic eigenvalues with no commutation or gap assumed (Theorem 16.5). Example 16.10 refutes extending that eigenvalue formula to arbitrary non-Hermitian coefficients; the abstract classification still covers them. The normal form and inhomogeneous existence rest on an imported splitting/surjectivity theorem. Part V (member 12):  $\partial y = -tAy$ ,  $t = \omega^{-1}$ , with ordinary complex residual matrix  $A_0$  and positive-order power-Hahn perturbation, classified over  $\mathbb{C}((t^{\mathbb{R}}))$  and over  $\mathbf{No}[i]$ ; the solution space has dimension the algebraic multiplicity of the real eigenvalues of  $A_0$ , and, with  $\lambda_j$  the eigenvalues of  $A_0$ ,  $\text{Ph}(-tA) = \{(\text{Im } \lambda_j) \log t\}$  is a non-Hermitian sector reached without that import. Part VI (member 11): first-order autonomous equations with constant coefficients on curves, classified (Theorem 30.1) with the same solutions for *every* normalized derivation (Theorem 31.4), no two asserted equal on  $\mathbf{No}[i]$ , and the logarithmic-derivative image of a constant abelian variety. *Merged; the hazard was a word.* “Phase” named the finite angle in  $\mathcal{O}$ , the infinitesimal residue in  $\mathfrak{m}$ , and the accumulated phase, no angle at all; one source printed two of them twenty lines apart. Each gets one symbol,  $\text{Ph}(A)$  collects obstructions of the third, member 12’s “phase labels” become residual frequencies,  $J$  is retired, and the meaningless merged sentence is printed beside the correct one. Later reproofs are cited as agreements; hedges that later members answer, some only in part, are amended in place, wording kept. Section 51 registers 106 non-claim items: the derivation is imported, not reproved, and no priority, refereeing or proof-assistant verification is claimed. Corollary 11.4 strengthens in phase-independence the prior chosen-exponential obstruction of [surcomplex/analysis](#);  $\text{Ph}$  is not the spectrum of [surcomplex/spectral-theory](#). Part VII (member 13): on the log-atomic tower,  $\partial^2 y + (Q_\alpha + c(\ell'_\alpha/\ell_\alpha)^2)y = 0$  has a two-dimensional solution space for  $c \leq 1/4$  and none for  $c > 1/4$  at every ordinal depth (Theorem 38.1); no set of earlier scales decides this (Theorem 39.2); at limit stages three nested Hahn fields give dimensions 0, 1, 2, with a Borel Galois group (Theorems 40.4, 41.2).

## 27. Rank-one non-Archimedean analytic geometry inside the surcomplex numbers

[surcomplex/rank-one-berkovich/article.pdf](#)

from [surcomplex\\_rank\\_one\\_berkovich.zip](#)

The report devoted to intrinsic rank-one nonarchimedean analytic convergence. Inside the rank-one workspace  $K = \mathbb{C}((t^{\mathbb{R}}))$  the absolute value  $|a|_v = \exp(-v(a))$  is genuine and  $K$  is spherically complete, complete and algebraically closed, so it is a legitimate base for Tate algebras and Berkovich spaces. Proves strict inclusions among the Tate, polynomial-coefficient Hahn, common-domain, radius-free germ, formal-coefficient and open-disk analytic rings with named witnesses, extending the ring table of [surcomplex/analytic-geometry](#); gives a constructive one-variable preparation theorem, principal-ideal and finite-quotient results, analytic root counts and a valuation



Rouché theorem; constructs the Berkovich disk points and the annulus retraction; and contrasts the contractible annulus spectrum with its nonzero analytic de Rham class. The worked case study is the entire series  $F(Z) = \sum_{n \geq 0} t^{n^2} Z^n$ , taken from [foundations-and-computation/foundations](#) and credited there: its full zero geometry is settled, and the first correction of the  $m$ th normalized root occurs at  $t^{m(m+1)}$ . *This does not contradict the no-convergence theorems of [surcomplex/analysis](#).* The convergence here is in the intrinsic valuation topology of one fixed rank-one field, not the fine topology induced from  $\mathbf{No}[i]$ ; and at  $Z = t^{-\omega}$  the exponents  $n^2 - n\omega$  strictly decrease, so the support is not well ordered and  $F$  is not an all-surcomplex entire function — the obstruction agrees with all-scale rigidity. The original audit credits the Berkovich *comparison* in [surcomplex/contours-and-stokes](#); its development-gap claim concerns that inspected snapshot, not an exhaustive search of the current collection.

## 28. Surcomplex spectral theory: singular values, infinitesimal rank, multiscale stability

[surcomplex/spectral-theory/article.pdf](#)

from three manuscripts, delivered 21 and 22 September 2026

Finite matrices over a real closed field  $F$  and  $F(i)$ , then over set-sized Hahn workspaces  $K_\Gamma = \mathbb{C}((t^\Gamma))$  inside  $\mathbf{No}[i]$ ; every dimension is an ordinary finite integer, and the analytic coefficient-ring machinery is not needed. The base manuscript proves the spectral theorem and the SVD *without compactness* (classical, not claimed as new): real closedness suffices, the witnesses are exhibited, and Courant–Fischer and Ky Fan extrema are *attained*; then positive square roots, polar decomposition, generalized Hermitian eigenproblems, Moore–Penrose inverses, least squares, best low-rank approximation and conditioning. Its core identifies singular-value valuations with successive differences of minimum minor valuations  $\Delta_k(A)$ , and recovers them and the leading singular coefficients from positive Gram coefficients one scale block at a time; applications include an intrinsic rank filtration, root-free scale elimination, precision certificates, nonnormal pseudospectra, gap-sensitive projectors and Schur-complement corrections. The second manuscript drops divisibility of  $\Gamma$ : Hermitian and normal descent and the SVD keep every exponent in the subgroup generated by the input supports, through a finite splitting tree; for positive semidefinite  $A$  and integer  $q \geq 2$  the principal  $q$ th root, and equally its trace, generates the field with exponent group  $\Gamma + \sum_{k=1}^{\text{rank } A} \mathbb{Z}\Delta_k(A)/q$ , of degree the order of the subgroup the  $\Delta_k(A)$  generate modulo  $q\Gamma$ ; at  $q = 2$ ,  $\Delta_k(A) \in 2\Gamma$  for  $1 \leq k \leq \text{rank } A$  is the exact Gram-factorization criterion. The third (Section 12) treats monic squarefree real-rooted  $p$  over  $F_\Gamma = \mathbb{R}((t^\Gamma))$ ,  $\Gamma$  arbitrary: with  $h_1, \dots, h_n$  the leading principal minors of its Newton-sum Hankel matrix and  $h_0 = 1$ , the splitting field is exactly  $F_\Gamma(\sqrt{h_1}, \dots, \sqrt{h_n})$ ,  $p$  has  $\#\{k : v(h_k) - v(h_{k-1}) \in 2\Gamma\}$  irreducible factors, and both persist for every monic  $\tilde{p}$  of degree  $n$  over  $F_\Gamma$  with  $v(h_k(\tilde{p}) - h_k(p)) > v(h_k(p))$  for all  $k$ ; applied to the squarefree part of  $\det(XB - A)/\det B$ , this gives the exact field of the generalized eigenvalues of a definite Hermitian pencil, not of a normalized eigenbasis. The pivot-class profile does not determine the factor degrees. Shared results are printed once, in the earlier manuscripts’ form; the third’s Hensel-lifting proof of real principal axes is kept as a marked second route, and its section sits here rather than in [surcomplex/polynomial-algebra](#), whose divisible workspaces make every such invariant vanish. Priority is not certified; not refereed or proof-assistant verified; the three repository audits were targeted and establish no absence elsewhere. [surcomplex/prony-reconstruction-at-surreal-scales](#) shares only the classical Hankel moment identity; [surcomplex/infinite-dimensional-hahn-spectral-theory](#) is the infinite-dimensional sibling, and no theorem of it is used here.

## 29. Surcomplex dynamics: linearization thresholds, periods, and normal forms

[surcomplex/dynamics-and-normal-forms/article.pdf](#)

from ten manuscripts, seven delivered 21 and three 22 September 2026

Local and global dynamics over complex Hahn fields  $K = \mathbb{C}((t^\Gamma))$  realized inside  $\mathbf{No}[i]$ , with  $\Gamma$  a set-sized ordered abelian group of *arbitrary* valuation rank. The report is the *union* of ten manuscripts: nine on germ and fixed-point dynamics, and one (§§17–19) on the quasi-periodic half, a cohomological equation on a real torus whose frequency vector has *surreal* components, with divisors  $k \cdot \Upsilon$  indexed by an integer lattice. **Read Section 2 first:** five inequivalent conditions travelled as *resonance*, six as *small divisor*, and two groups of sources used *radius* with **opposite** monotonicity; each variant is named, every threshold states its variant and coefficient category, and Section 2.5 is a tag ledger. Lattice resonance is *not* frequency resonance, and visible divisor valuations are *not* a spectrum. Quasi-periodic results:  $d$  surreal components give *at most*  $d$  arithmetically visible divisor valuations, via a canonical descending flag of saturated integer resonance lattices of strictly dropping rank, with one well-ordered set containing every reciprocal nonzero divisor’s support (not a summability claim); an exact stratified subexponential criterion for universal solvability on one fixed torus strip; an arbitrary-order lifting obstruction; and, with no lattice resonance, under that criterion and for perturbations of valuation above the loss  $\kappa_\Upsilon$ , a unique normalized mean-zero conjugacy to a constant vector field with a frequency correction, by marked support words rather than sequential valuation convergence. Germ results: the exact maximal centered valuation ball read off the finite  $q$ th return iterate; a five-row coefficient-category table, necessity witnessed per row at one exponent; an algebraicity dichotomy over  $\mathbb{C}(w)((t^\Gamma))$  with monodromy and finite-cover obstruction; periods as a complete conjugacy invariant with explicit moduli; a gauge-normalized symplectic normal form with no arithmetic bound. Sources 08 and 09 (§7; source 08 also §8.6) answer the common-domain question for an *exact scalar* multiplier, oppositely to source 02’s drifted answer. For  $\lambda I_d$ ,  $\lambda = e^{2\pi i\theta}$  with  $\theta$  irrational, any nonzero  $\Gamma$ ,  $d \geq 1$ ,  $\tau < \infty$  and input coefficients  $f_s$  holomorphic on one polydisk of radius  $R$  with  $f_s(0) = 0$ ,  $Df_s(0) = 0$ , all coefficients of the normalized conjugacy and its inverse are holomorphic on the open polydisk of optimal universal radius  $Re^{-\tau}$  (Theorem 7.13, credited to both). Under *drift*, beyond source 02’s  $\text{rad}(h_{k,1}) = R \exp(-(k+1)\sigma)$  and its trichotomy, source 08 shows that one Hahn monomial over  $\mathbb{Z}$  already gives radii exactly  $R \exp(-m\sigma)$  for  $0 < \sigma < \infty$  (Theorem 8.15); in one variable over any fixed nonzero  $\Gamma$ , every input’s normalized linearizer has a common positive coefficient disk iff  $\sigma = 0$  (Corollary 8.17). Source 08 is the base (every  $d$ , and the drifted theorem); source 09, translated from its forward convention (its  $H$  is the report’s  $H^{-1}$ ), adds the tree product bound (Theorem 7.5) and a plane-tree proof for  $d = 1$ , kept as a marked second route sharing the arithmetic lemmas. Source 10 (§§7.7–7.13) settles the nonscalar exact case: every exact nonresonant diagonal unitary  $\Lambda$  with  $\tau < \infty$  has a common polydisk of radius  $Re^{-r\tau}$ ,  $r$  the rational rank of the angles (Theorem 7.23), sharp when  $r = 1$ ; the optimal radius for  $r \geq 2$  stays open (Question 25.1). Section 26 keeps all 123 source limitations. Extends [surcomplex/differential-equations](#), but uses external coordinate derivatives, not its Berarducci–Mantova derivation; kept apart from the disks of [surcomplex/rank-one-berkovich](#). Not refereed; priority not certified; Lean checks only the support lemma, the combinatorial ancestry bounds of Corollary 3.5 (corrected to keep the input support) and the counterexample to an input-independent bound, not that corollary’s operator-summability assertion.

## 30. Nonabelian support obstructions: a matrix Cousin criterion and the surcomplex Riemann–Hilbert problem

[surcomplex/nonabelian-support/article.pdf](#)

from three manuscripts, two delivered 21 September 2026 and one 22 September 2026

One coefficient category, studied from two sides and then past its integrality boundary: matrices  $I_r + \text{Mat}_r(J_\Gamma)$ , congruent to the identity modulo positive Hahn exponents over a set-sized, possibly non-Archimedean value group  $\Gamma$ , with ordinary holomorphic coefficients on a *single fixed* ordinary complex domain ( $t^\gamma = \omega^{-\gamma}$  reads them in  $\mathbf{No}[i]$  when  $\Gamma \subseteq \mathbf{No}$ ). Part I solves a matrix Cousin problem on a puncture-and-disk cover of a connected open Riemann surface,

with integral vector bundles and a Stein branch; Part II inverts monodromy for near-identity representations of  $\pi_1$  of a plane minus a closed discrete set, conversely to the forward map of **surcomplex/differential-equations**. **Both headline theorems read “the union of the supports is well ordered” and are *not* the same criterion:** gluing (Theorem 6.2) tests the *normalized polar factors*  $\text{Pol}(G_a) - I_r$ , each transition factored independently and first, and well-ordering of full raw transition support is sufficient but not necessary, whereas raw singular-coefficient support fails in *both* directions. Realizability (Theorem 14.2) tests the *raw* monodromy matrices  $\rho(\ell_d) - I_r$ , whose support has basis- and basepoint-independent well-ordering. The trap is printed: a determinant-one rank-three family over  $\Gamma = \mathbb{Q}$  with one raw singular exponent and a *splitting* abelianized cocycle admits no splitting; one added term putting the forced descending set among the raw poles *removes* the obstruction; the true matrix logarithm is no repair, one coefficient off by a factor  $1/2$ . Also: rank-one recovery (cited, not reproved) of the scalar Cousin criterion of **surcomplex/global-divisors**, the right torsor of all splittings, absoluteness under ordered value-group extension, and for every  $r \geq 3$  a cocycle trivial modulo the last central subgroup yet nonsplit in full positive rank. Part III (sections 19–27), from the third manuscript, glues by finite exponentials  $\exp(t^{\alpha_n} e^{1/(z-n)} N)$ ,  $N$  nilpotent,  $\alpha_1 > \alpha_2 > \dots > 0$ , putting descending exponents before ordinary *essential* singularities: determinant-one plane bundles, integral-trivial on bounded domains, whose Jordan-block members of rank  $r \geq 2$  stay indecomposable and not stably trivial over  $\mathcal{O}((t^\Gamma))$  and even over  $\text{Mer}((t^\Gamma))$ . It computes all sections and morphisms, classifies this one family (for fixed  $r \geq 2$ : profiles eventually equal integrally, eventually differing by one constant  $\delta \in \Gamma$  after localization), and proves a localization-stable cyclic dichotomy, continuum many classes in every noncyclic group and a tensor rule; the same manuscript supplies Theorem 28.4: for rational binary profiles neither isomorphism nor nonisomorphism is semidecidable. It does *not* settle Part I’s printed localization question: a merge observation (Corollary 27.2) makes Part I’s pole-data examples and the global-divisor report’s explicit plane Picard classes trivial over  $\text{Mer}((t^\Gamma))$ , deciding nothing over  $\mathcal{O}((t^\Gamma))$ . Shared lemmas are printed once; Part III’s labels are `nab:loc:`, the 169 earlier `nab:` labels intact. Not refereed, no Lean code, no priority certified.

### 31. Cofinality, factorization and scalar extension for entire Hahn functions at arbitrary rank

[surcomplex/entire-functions-at-arbitrary-rank/article.pdf](#)

from six manuscripts: two delivered 21 September 2026, four 22 September 2026

Ordinary power series  $\sum a_n Z^n$  over a *fixed* Hahn field  $K_\Gamma = \mathbb{C}((t^\Gamma))$ ,  $\Gamma$  nonzero, set-sized, divisible and of arbitrary valuation rank; *entire* means strongly Hahn-summable at every point of that one field; for  $\Gamma = 0$  the ring is  $\mathbb{C}[Z]$ . The first two sources, which disclaim all of **No**[ $i$ ], proved the same engine, printed once; both payloads are kept. **The trichotomy** (Theorem 1.5):  $E_\Gamma$  is always a GCD domain with units  $K_\Gamma^\times$ ; a polynomial PID when  $\text{cf}(\Gamma) > \aleph_0$ ; non-Noetherian Bézout with full Hermite interpolation when an *order unit* exists, sharply so (Theorem 4.1); **not** Bézout when  $\text{cf}(\Gamma) = \aleph_0$  without an order unit, two canonical products with disjoint simple zero sets and gcd 1 generating a proper ideal. **The exact jet image** (Theorem 1.6), from the third source, answering the question the first merge left open: jets on an infinite radially finite divisor  $D$  are realized iff on all late shells the shell’s Hermite polynomial has coefficients in the ring  $V_n$  that kills the Archimedean scale of the shell radius; so  $E_\Gamma/(P_D)$  is an *algebraic* restricted product, not a topological or adelic one, with a sharp Bézout criterion. The fourth source, pinned before the third was merged, is recorded as an independent second proof of these theorems, its resolution claim stale; it adds finite-generator criteria, membership for one simple node per shell, the exact threshold for paired zeros, and, for one node per shell, all maximal ideals containing  $P_D$ : the evaluations and one per nonprincipal ultrafilter, a copy of  $\beta\mathbb{N}$ . None of this is a corona theorem; several nodes per shell and prime ideals stay open. **Scalar extension** (Theorem 1.7): for divisible  $\Gamma \subseteq \Delta$ , all nonpolynomial entire series share one exact strong evaluation domain in  $K_\Delta$ , the valuation ring coarsened by the convex hull  $H$  of  $\Gamma$ ; entireness survives exactly for cofinal extensions, no new zeros appear, and “not entire over  $K_\Delta$ ” is the coarse form of that exact domain. The fifth source (Section 11), without divisibility and over any trivially valued coefficient field, answers only the scalar-extension

clause of the several-variable question: a whole-family entire series is strongly summable at a point with nonzero coordinates iff the weight image of its ordinary support is well ordered in  $\Delta/H$ ; under homogeneous-block summation, never called entire, the polynomial-direction loci of nonpolynomial entire series are, when  $\text{cf}(\Gamma) = \aleph_0$ , exactly the proper countable unions of projective algebraic sets. Extends [surcomplex/rank-one-berkovich](#), which lands in the Bézout case; the whole-class rigidity re-derives, with credit, a conclusion already in [surcomplex/analysis](#) and [foundations-and-computation/foundations](#); the restricted product here is not that of [surcomplex/global-divisors](#). Rank is not the driver:  $\bigoplus_{\alpha < \omega_1} \mathbb{Q}e_\alpha$ , ordered by the largest nonzero index, admits only polynomials. No source was refereed or machine-checked, and priority is not certified. A sixth manuscript (Section 12) classifies every prime ideal above a canonical product with one simple node per shell: the quotient is zero-dimensional exactly when  $\Gamma$  has an order unit, otherwise continuum chains of primes appear, and a cofinal group extension can split one prime into a continuum-sized chain while the zeros are unchanged.

### 32. Holonomic rigidity for entire Hahn functions: a sharp order-unit criterion for dilations

[surcomplex/holonomic-rigidity-for-entire-hahn-functions/article.pdf](#)

from five manuscripts, delivered 22 September 2026

Power series over one fixed Hahn field  $K_\Gamma = \mathbb{C}((t^\Gamma))$ , with  $\Gamma$  a nonzero set-sized ordered abelian group *not assumed divisible*; *entire* means that the evaluation family is strongly Hahn summable at every point of that one field, never of all of  $\mathbf{No}[i]$ . Theorem A: a nonzero linear differential operator  $L$  has  $B_L \geq 0$  and  $N_L$ , determined by its finitely many coefficients, such that a formal solution strongly evaluable at one  $x \neq 0$  with  $v(x) \leq -B_L$  has degree below  $N_L$ ; so entire D-finite series are polynomials, also in  $d$  variables (Theorem C). Theorem B, the principal proposed contribution: for  $q$  of infinite multiplicative order, a nonpolynomial entire solution of a nonzero polynomial-coefficient linear  $q$ -difference equation exists **iff**  $v(q) \neq 0$  and  $|v(q)|$  is an order unit of  $\Gamma$ ; units with root-of-unity residue are covered, and a classical partial theta series is the witness. The engine is a finite-step escape chain forcing nonincreasing leading exponents, so the recurrence exclusion region is closed, and a formal exponential attains it (Proposition 5.3), while the coarsened region of Theorem 7.4 must be open. Also the exact partial theta domain for any Hahn parameter  $p$  with  $v(p) > 0$  (Theorem 8.1), mixed differential–dilation rigidity at nonzero noncofinal scales (Theorem 10.2), and, for  $\Gamma = \bigoplus_{j \geq 0} \mathbb{Q}\omega^j$  (no order unit), the entire series  $\sum_n \omega^{-\omega^n} z^n$ , satisfying no nonzero linear differential equation, nor a nonzero linear dilation equation for any nontorsion parameter (Theorem 11.2). Source 08, assuming only  $\Gamma \neq 0$ , is the base; proofs from 09, which assumes divisibility, were re-read for hidden divisions. Divisibility is now restricted to three threshold constructions (Section 1.4): inward stability of strong evaluation removes it from the torsion eigenspace equivalence in Proposition 9.1. The recurrence thresholds handle an empty forward set explicitly, and coarsening changes the valuation on the field rather than projecting every Hahn support to a smaller exponent group. Remarks 4.5 and 8.2 keep both sources’ proofs, and both exact-arithmetic suites are kept (08: 3,705 checks; 09: 2,043; no failures; no infinite statement proved). Not claimed: torsion  $q$  (counterexamples in Section 9), nonlinear differential or unit-valuation mixed equations (open, Questions 15.1–15.2), optimal thresholds, a named conjecture, certified priority or refereeing. The Lean ledger maps the positive-support word lemma, escape mechanism, bounded-cost exclusion and formal exponential domain; the main classifications remain unformalized.  $D_z$  kills every scalar: it is not the Berarducci–Mantova scalar derivation used in [surcomplex/differential-equations](#). From [surcomplex/entire-functions-at-arbitrary-rank](#) it cites the coarsening lemma and background, not the ring theorems, and that report’s order-unit dichotomy (Bézout, Hermite interpolation) is **not** this one. As [surcomplex/rank-one-berkovich](#) has  $\sum t^{n^2} z^n$ , the theta witness is not counted as new. A third manuscript (Sections 14–17) proves that every strongly entire solution of a nonzero *first-order algebraic* differential equation over  $k((t^\Gamma))$ , any field  $k$  of characteristic zero and any  $\Gamma \neq 0$ , is a polynomial (Theorem D), with a higher-order corner criterion and Euler rigidity; higher-order nonlinear equations stay open (Question 19.1), and the



wider coefficient field is not transferred to Theorems A–C. Two later manuscripts (Sections 20–22) answer the nonlinear question when  $\Gamma$  has no order unit: a differential or dilation relation forces all Taylor coefficients into one finitely generated field, so every nonpolynomial strongly entire series is differentially transcendental in all orders, jointly with its non-torsion dilates, and a continuum-sized differentially independent family exists; the order-unit case stays open.

### 33. Hahn–Tate uniformization at arbitrary valuation rank

[surcomplex/hahn-tate-uniformization/article.pdf](#)

from three manuscripts, delivered 22 September 2026

Tate elliptic curves over  $K = \mathbb{C}((t^\Gamma))$ , with  $\Gamma$  set-sized, of any rank and not necessarily divisible,  $q \in K$  of valuation  $\alpha > 0$ ,  $H_\alpha$  the convex subgroup of values bounded by an ordinary integer multiple of  $\alpha$ , and  $U_q = v^{-1}(H_\alpha)$ . **Theorem 1.2:** the bilateral theta family, and for  $u \in K^\times \setminus q^\mathbb{Z}$  each Tate coordinate family, is strongly Hahn-summable *exactly* on  $U_q$  — outside it the leading exponents contain a strictly decreasing sequence, so no cancellation repairs it — yet  $\Phi_q : U_q \rightarrow E_q(K)$  is a surjective homomorphism with kernel  $q^\mathbb{Z}$ . Source 01 is the base; source 03, written without knowledge of it, proved the theorem by another route, and both proofs are kept: Tate’s rank-one theorem in  $K_{H_\alpha}$  with coarse lifting (Sections 5–7), and an integral node chart with  $UV = q$  plus two further formal charts, with no rank-one completion (Section 8), valid over any coefficient field of characteristic zero (Theorem 8.13), hence over **No** for real  $q$  (Corollary 13.2). The domain half, the value sequence  $1 \rightarrow \mathcal{O}_v^\times \rightarrow E_q(K) \rightarrow H_\alpha/\mathbb{Z}\alpha \rightarrow 0$ , the good coarse reduction and the  $q = \omega^{-1}$  example are printed once. Also: exact support and valuation identities in the node chart; if  $H_\alpha \neq \Gamma$ , no homomorphism  $K^\times \rightarrow E_q(K)$  extending  $\Phi_q$  has kernel  $q^\mathbb{Z}$  (Theorem 10.5); recovery of  $q$  from  $j$  when  $v(j) < 0$  over any characteristic-zero coefficient field; and, over  $\mathbb{C}$ ,  $K(E_q[n]) = K_{\Gamma+\mathbb{Z}\alpha/n}$ . Every elliptic curve over **No**[ $i$ ] with  $v(j) < 0$  is isomorphic to an  $E_q$ . Part II, source 02, assumes monomial periods, coefficients in  $\mathbb{C}$  and  $\Gamma$  divisible where halves are used: every admissible lattice in a finite-dimensional higher-rank inner-product space has a finite positively generating set (Theorem 15.2), answering affirmatively Question 11.14 of Amini–Nicolussi *as printed* in 2025; an exact flag criterion for quadratic lattice families, support order type  $\omega^s$ , the exact multiscale theta domain, finite minimum certificates, and lifting of smooth initial zeros. **The bounded-period quotient is not new:** Molcho–Wise’s bounded monodromy and logarithmic Tate curve are a precedent, correcting source 01’s novelty framing, and no functorial comparison is proved. No abelian variety is uniformized,  $v(j) \geq 0$  is not treated; not refereed, no Lean verification, priority not certified by the targeted searches. Neighbours: [surcomplex/rank-one-berkovich](#) (Tate *algebras*, not Tate curves) and, for Part II, [surcomplex/wick-summability-certificates](#).

### 34. Infinite-dimensional Hahn spectral theory: infinitesimal thickening, range defects, and algebra-relative spectra

[surcomplex/infinite-dimensional-hahn-spectral-theory/article.pdf](#)

from four manuscripts, delivered 22 September 2026

Two manuscripts over one Hahn field  $K = \mathbb{C}((t^\Gamma))$ , printed as *two parts*: *spectrum* means different things on their two vector spaces and in their different operator algebras. Section 3 sets the two side by side on  $\text{diag}(1/n)$ ; each part is self-contained given Section 2. **Part I:**  $H$  an ordinary complex Hilbert space,  $\Gamma$  nonzero, set-sized and ordered abelian, without divisibility (only Section 12.1 assumes  $\Gamma \subseteq \mathbb{R}$ ), and  $H((t^\Gamma))$  with the coefficientwise-convolution inner product. Every everywhere-defined adjointable operator, with no continuity assumed, is a Hahn series of ordinary bounded operators with one common well-ordered support (Theorem 6.1). For ordinary bounded normal  $T$  the failure-of-bijectivity spectrum is  $\sigma_\mathbb{C}(T) \cup (\sigma_\mathbb{C}(T)' + \mathfrak{m}_K)$ , with no new eigenvalues (Theorems 8.2, 8.3): accumulation points thicken into infinitesimal monads, isolated points do not. An injective linear  $S : V \rightarrow V$  keeps  $\text{coker}(S+E) \cong W((t^\Gamma))$ , where  $V = \text{Ran } S \oplus W$ , under every  $E$  of strictly positive order (Theorem 7.1). With  $De_n = e_n/n$  on  $\ell^2$  and  $\eta > 0$ ,  $C = D^2 + t^{2\eta}I$  is self-adjoint and coercive with the infinitesimal constant  $t^{2\eta}$ , yet its cokernel has dimension at least  $2^{\aleph_0}$  (Theorem 11.3), and at every rank its range is closed and proper with



zero orthogonal complement (Theorem 12.2). A least field-valued norm bound exists exactly when  $T$  attains its ordinary norm (Theorem 13.2). **Part II** also allows arbitrary ordered value groups. In  $\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$  acting on  $\mathbb{C}^{(I)}((t^\Gamma))$ ,  $A = \text{diag}(d_i) + B$ , with pairwise distinct ordinary real  $d_i$  and  $B = B^*$  of globally well-ordered positive support, has a unique near-identity normalized Hahn-unitary diagonalization with corrections in the monoid generated by  $\text{supp } B$  (Theorem 18.2, Corollary 18.3), exact synthesis, classified commutant and spectral projections, and  $\sigma_{\mathcal{A}_I}(A) = \{\lambda_i\}$  even when the  $d_i$  accumulate (Theorem 20.2) — a spectrum relative to that algebra, not a Banach-algebra spectrum. Closed weighted walks control every eigenvalue coefficient, and the one-scale finite-section bound  $(2r + 2)\eta$  (Corollary 21.5) cannot be improved in general (Theorem 21.6). *Merged:* the Hahn–Neumann support lemma is stated once, and Part II’s proof of Higman’s lemma is not reprinted, the identical argument being `spec:lem:words` and `spec:lem:neumann` of `surcomplex/spectral-theory`. The proof review removes the unused divisibility assumption of Part I and proves range closedness by its continuous defect projection at every rank; neither part uses a theorem of the other. It supplies the first two items of that report’s closing non-claim and the compact half of the third, compact meaning compact on  $H$ . Part II supplies a vectorwise Hahn-additive projection calculus, with a rank-one counterexample to pointwise valuation convergence. No trace-class theory or general topological spectral-measure theorem is provided; `surcomplex/hahn-herglotz-positivity` gives a separate obstruction to a positive coefficientwise circle measure. Both suites report 218 checks, by coincidence; the assertions differ. Priority not certified; not refereed; no proof-assistant check. Two later manuscripts add Parts III and IV. Part III computes the spectrum of a constant element relative to  $\mathfrak{A}((t^\Gamma))$  for any unital algebra  $\mathfrak{A}$  through Drazin invertibility (Theorem 32.1): it is Part I’s adjointable spectrum for  $B(H)$  and Part II’s for row- and column-finite matrices, and recovers only that half of Part I’s theorem. Part IV, with  $\Gamma$  divisible, treats compact self-adjoint operators under noncommuting positive-valuation perturbations: finite Hermitian clusters at nonzero eigenvalues, the whole monad of zero in the spectrum, and no global bounded-coefficient spectral theorem; its coefficientwise Fredholm determinant is not a trace-class theory.

### 35. Expanding polynomial dynamics over surreal and surcomplex fields: exact symbolic fibers and the order-unit obstruction

`surcomplex/expanding-polynomial-dynamics/article.pdf`

from one manuscript, delivered 22 September 2026

Iteration of  $F = q^{-1}P$  over  $K = k((t^\Gamma))$ ,  $k = \mathbb{R}$  or  $\mathbb{C}$ , with  $\Gamma \neq 0$  any set-sized ordered abelian group — no divisibility, no algebraic closedness — where  $\kappa = v(q) > 0$ ,  $P$  is integral of degree  $d \geq 2$ , and its reduction keeps degree  $d$  and has  $d$  distinct roots in  $k$ . With  $I_\kappa = \{h : v(h) > n\kappa \text{ for all } n \in \mathbb{N}\}$ : every word  $s \in \{1, \dots, d\}^\mathbb{N}$  has a canonical center  $x_s$  with  $F(x_s) = x_{\sigma s}$ , built by formal algebra and strong Hahn evaluation under one support certificate for all words; the integral nonescaping set is  $\bigsqcup_s (x_s + I_\kappa)$ ; and words first differing at index  $m$  give points at valuation distance exactly  $m\kappa$ . So a word determines one point *iff*  $\kappa$  is an order unit — a property of the scale, not of the rank. The preperiodic points are exactly the centers of eventually periodic words, and each  $F^{m+p} - F^m$  has  $d^{m+p}$  simple roots in  $K$ . **Order-unit dichotomy:** for an order unit,  $s \mapsto x_s$  is a homeomorphism from the full shift onto a compact perfect set; otherwise the fibers are open and the iterates uniformly equicontinuous, though the Hausdorff quotient of the  $\kappa$ -scale uniformity is still the full shift, and a separate compact-subsystem rigidity theorem makes every compact forward-invariant subset finite. Also scalar extension (fibers can thicken, no new preperiodic points), deformations with exact leading response, the exact valuation-bounded locus, a no-order-unit equicontinuity criterion for all polynomials, and  $F(x) = \omega(x^2 - 1)$  in  $\mathbf{No}[i]$  with real centers. One source, not a merge: nothing was selected, and at placement only terminology changed, and a later proof review clarified degree preservation, formal substitution and topology. The manuscript’s “cofinal scale” is an order unit here, since `surcomplex/entire-functions-at-arbitrary-rank` reserves cofinality for  $\text{cf}(\Gamma)$ . Valuation-expanding is not repelling and no Julia/Fatou theory is built, so neither exclusion of `surcomplex/dynamics-and-normal-forms` is answered. In  $\mathbf{No}[i]$  the set-level discreteness and compactness clauses are special cases of the closed-and-discrete theorem

of `surcomplex/analysis`; the substance lies in set-sized workspaces. The order-unit case has the classical symbolic Cantor model even at higher rank; every positive scale is an order unit in the field of `surcomplex/rank-one-berkovich`;  $H_\kappa$  is the  $H_\alpha$  of `surcomplex/hahn-tate-uniformization`, a shared subgroup, not a shared theorem. Repeated residue roots and rational maps are not covered. Not refereed; priority not certified. Lean proves the scale subgroup and error-ideal lemma, including the valuation-ball conditions for clopenness and the coarsened valuation ring. The dynamical theorems remain pending; the 38,139 exact finite checks prove no infinite statement.

### 36. Positivity without a positive measure: Hahn–Herglotz normalization and strict moment hierarchies in surcomplex analysis

`surcomplex/hahn-herglotz-positivity/article.pdf`

from one manuscript, delivered 22 September 2026

Over  $\mathbb{R}((t^\Gamma))$  and  $\mathbb{C}((t^\Gamma))$ , with  $\Gamma$  a nonzero set-sized ordered abelian group, and by transport over set-sized subfields of  $\mathbf{No}[i]$ , finite Toeplitz positivity, positive real part on the halo of an ordinary domain, and positive-measure representation on the *ordinary* circle come apart. Measures are real-valued and *coefficientwise*: an ordinary finite signed regular Borel measure at each exponent of one well-ordered support, additive exponent by exponent, without requiring strong summability; moments are Fourier moments. For divisible  $\Gamma$ , assumed only for the congruence in Theorem 3.3, a matrix coherent function with  $\operatorname{Re} H \succeq 0$  at ordinary points is, after subtracting a constant  $iB$  with  $B$  Hermitian, constant-congruent to  $G \oplus 0$  with  $G(a_0) = I$ ,  $\operatorname{Re} G_0 \succ 0$  and otherwise unrestricted higher coefficients. For every nonzero  $\Gamma$ , ordinary and halo positivity agree, kernels are constant, and Harnack holds with every ordinary  $C > C_0$  but need not with the sharp classical constant (Theorem 3.5). Theorem 4.2:  $\sum \nu_\gamma t^\gamma$  is positive iff each  $\nu_\gamma^-$  vanishes on the common null ideal of the earlier  $|\nu_\beta|$ . Corollary 5.2 gives a strictly positive, unital, pointwise completely positive functional with no positive representing measure. The moments  $c_0 = 1$ ,  $c_n = -\varepsilon$  have  $\det T_N = (1 + \varepsilon)^N(1 - N\varepsilon) > 0$  for every ordinary  $N$ , yet their unique signed measure is  $(1 + \varepsilon)m - \varepsilon\delta_1$ , possible only because  $\varepsilon$  is infinitesimal; positive quadratures converge coefficientwise weak-\* to it, the Schur parameters are  $-\varepsilon/(1 - n\varepsilon)$ , and the Schur iterates have internal poles of standard part 1. Theorem 7.1 separates the positive-measure, signed-measure, distribution, Toeplitz-only and halo-Herglotz classes strictly, with exponents  $0, \eta$  only. A single source; the main equivalence of Theorem 4.2 is proved once, in `surreal/hahn-valued-measures-and-probability` as `meas:thm:nullideal`, found independently by a source of that report in the same batch; this report proves the translation between the two forms and the control-measure reformulation, and keeps its source’s uncountable example showing that the latter cannot replace the null ideal. In that report’s strong class  $\omega + 1$  is the least support order type letting nonnegative finite tests hide a negative event; in the coefficientwise class here two exponents suffice. The halo is that of `surcomplex/analysis`, whose `c:p5:positivity` is the mechanism of Theorem 5.1(b); Corollary 6.5 adds only negative information to `surcomplex/spectral-theory` and `surcomplex/infinite-dimensional-hahn-spectral-theory`. Among 26 non-claims: nothing on enlarged circles, finitely additive or nonstandard representations, or a matrix null-ideal criterion. A candidate contribution; priority not certified and not refereed. Lean proves the null-ideal and two-scale criteria and finite algebraic clauses of the negative-atom, quadrature and Schur examples; the remaining analytic and representation claims are pending.

### 37. Sharp moment reconstruction at surreal scales: collision geometry, optimal precision, and Hahn-supported inversion

`surcomplex/prony-reconstruction-at-surreal-scales/article.pdf`

from one manuscript, delivered 22 September 2026

Recovery of  $n$  distinct integral nodes  $a_i$  and nonzero weights  $w_i$ , of any valuation, from the power moments  $m_k = \sum_i w_i a_i^k$ ,  $0 \leq k < 2n$ , over  $\mathbb{R}((t^\Gamma))$  or  $\mathbb{C}((t^\Gamma))$  with  $\Gamma$  any nontrivial ordered abelian group, not assumed divisible. No measure or positivity is involved; since neighbouring reports use “moment” otherwise, the directory is named after Prony. With  $d_i = \sum_{j \neq i} v(a_i - a_j)$ ,  $\delta_i = \max_{j \neq i} v(a_i - a_j)$ ,  $E_i = v(w_i) + 2d_i$  and  $\Theta = \max_i (E_i + \delta_i)$ , Theorem 1.1: if all  $2n$  moment

errors have valuation at least  $\kappa > \Theta$ , there is a unique regular  $n$ -node realization, uniquely labelled by the strict nearest-neighbour balls  $v(\hat{a}_i - a_i) > \delta_i$ , with node errors of valuation  $\geq \kappa - E_i$  and weight errors  $\geq \kappa - 2d_i - \delta_i > v(w_i)$ ; perturbing only  $m_{2n-1}$  attains every node bound at once, and at valuation exactly  $\Theta$  leaves no regular realization with that labelling, for *every* configuration. Optimality is for this labelling, not for bare existence; failure at  $\Theta$  need not be a collision or a loss of positivity. Also: exact Gauss valuations  $-2d_i$  and  $-2d_i - q_i$  of the inverse Hermite rows, with a cancellation depth  $q_i$  that the separation tree does not determine; the  $q_i$ -improved weight bound, proved for the tangent map, and for the whole nonlinear set only under the extra finite criterion (not claimed optimal) of Proposition 7.2, an arbitrary-rank form of the Caruso–Roe–Vaccon precision-lattice principle, with Proposition 9.1 showing that the improvement can fail nonlinearly; a *sufficient* directed-path certificate for nonuniform precisions, valid also for nondivisible  $\Gamma$  by auxiliary scaling in its divisible hull and descent to the original field; and transfer to finite configurations in **No** and **No**[ $i$ ]. One manuscript, not a merge; the only choices were editorial (title kept, labels prefixed **prony**:, notation aligned with **surcomplex/polynomial-algebra**). Remark 6.2, added on placement and proving no new theorem, compares  $\Theta = \max_i((v(w_i) + d_i) + (d_i + \delta_i))$  with that report’s coefficient threshold  $T(P) = \max_i(d_i + \delta_i)$ . Only the classical Hankel identity is shared with **surcomplex/spectral-theory**; the Prony-type lemma of **surreal/hahn-valued-measures-and-probability** is cross-referenced, not merged. Classical Prony, Hermite and precision-tracking material is credited, as are the  $w_i P'(a_i)^2$  denominator and the last-moment direction. Not an unconditional algorithm on surreal inputs; confluent and infinite configurations are out of scope; the 10,052 exact SymPy assertions check finite instances only. Priority is not certified; not refereed. The Lean ledger records classical uniqueness, cofactor orthogonality, the exact last-moment annihilator and algebraic local-root identities; valuation localization and the remaining precision estimates are pending.

### 38. Finite certificates for surcomplex Wick summability: valuation balancing, connected obstructions, and multiscale Gaussian perturbations

[surcomplex/wick-summability-certificates/article.pdf](#)

from one manuscript, delivered 22 September 2026

Formal Gaussian perturbation theory over  $K = \mathbb{C}((t^\Gamma))$ ,  $\Gamma \neq 0$  set-sized, *divisible* and of arbitrary rank: a finite interaction  $P = \sum_a g_a x^{\alpha^a}$ ,  $g_a \in K^\times$ ,  $\alpha^a \in \mathbb{N}^d \setminus \{0\}$ , and a symmetric *bilinear* covariance  $C$ , possibly singular and not necessarily real or positive; the expectation is a finite sum over pairings. With  $\lambda_a = v(g_a)$  and  $c_{ij} = v(C_{ij})$  on nonzero entries, Theorem 6.1: the vacuum Wick diagram family is strongly summable iff its weight is positive on the finite Hilbert basis of  $S = \{(m, k) : Am = Dk\}$ , iff some  $p \in \Gamma^d$  has  $\lambda_a + \alpha^a \cdot p > 0$  and  $c_{ij} - p_i - p_j > 0$  (a diagonal monomial rescaling making all couplings and nonzero covariances infinitesimal), iff every observable sector, equivalently any one pairable sector, is summable; Hahn-unit tails are arbitrary. The bridge is a strict alternative over any divisible ordered group, by Fourier–Motzkin elimination without real separation (Theorem 5.1); failure yields a nonnegative integer certificate in  $S \setminus \{0\}$  of nonpositive weight, whose multiples obstruct. With every valence at least 2 the connected vacuum diagrams have the same domain (Theorem 7.3; valence 1 fails, Example 7.5). Positive-definite real covariance, or diagonal covariance with nonzero diagonal, reduces the test to  $\lambda_a + \frac{1}{2} \sum_i \alpha_i^a c_{ii} > 0$  per monomial (Theorem 8.2), giving a sharp multiscale stationary-phase threshold (Corollary 8.3); in a mixed quartic each interaction alone is admissible but the pair is not. **Diagramwise only**: for  $C = I_2$ ,  $P = t^{-1}(x + iy)^4$  the order-grouped sum is 1 while the diagram family is not summable, and the criterion is not invariant under general complex linear coordinate changes. A Hahn value is not an integral, summability is not convergence of truncations in rank two (Example 9.4), and no physics is claimed. Exact solver over lexicographic  $\mathbb{Q}^r$ ; 7,684 finite checks at ranks 1–3. Single source, not a merge; placement added Sections 1.3, 13.4 and 13.5, and Section 13.1 corrects the source’s report that a repository search for *Gaussian* found nothing. Where **physics/surreal-scalars-and-spacetime** cautions that admissibility is not automatic, this report decides it exactly for one class, and answers none of that report’s questions; the closest relative is the multiscale theta series of **surcomplex/hahn-tate-uniformization**, also decided by

a finite certificate but by different mathematics; neither uses the other. Classical inputs credited; priority provisional; not refereed. The Lean ledger covers the positive-support, finite-semigroup and valuation Cauchy–Schwarz lemmas; the main summability and connected-domain theorems remain pending.

### 39. Three duals at surreal scales: completion, strong maps and valuation continuity

[surcomplex/three-duals-of-hahn-vector-spaces/article.pdf](#)

from one manuscript, delivered 22 September 2026

For a  $k$ -vector space  $V$  and  $V_\Gamma = V((t^\Gamma))$  over  $k((t^\Gamma))$ , three objects are kept apart: the algebraic scalar extension, its valuation completion, and its closure under strong Hahn summation. The completion consists exactly of the series whose coefficients below each valuation cut span a finite-dimensional space; in infinite coefficient dimension it is the whole Hahn space exactly when  $\Gamma \cong \mathbb{Z}$ , and it is strictly larger than the algebraic extension exactly when  $\Gamma$  has countable cofinality; a cofinal exponent enlargement preserves this and a noncofinal one collapses it. Every strong linear operator is one Hahn series of arbitrary  $k$ -linear coefficient maps. For infinite-dimensional Hilbert  $H$  and noncyclic  $\Gamma$  there are three strictly nested duals — inner-product-represented, strong, valuation-continuous — the first gap being the Hahn extension of  $H^\# / H'$ ; all strong maps are bounded by a positive real Hahn scalar; and a choice-based hyperplane has a strong bounded projection but zero orthogonal complement. It works on the Hahn–Hilbert space of [surcomplex/infinite-dimensional-hahn-spectral-theory](#), whose non-represented functional it shows to be strong, and whose adjointable operators sit inside its strong ones. One manuscript; some cited literature was not available to its author and is attributed, not verified; not refereed, no Lean verification.

### 40. Hidden negative directions of Hermitian forms

[surcomplex/hidden-negative-hermitian-directions/article.pdf](#)

from one manuscript, delivered 22 September 2026

For divisible  $\Gamma \neq 0$  let  $\mathcal{P}_\Gamma \subset \mathbb{C}((t^\Gamma))$  be the series supported in finitely generated subgroups: an algebraically closed subfield containing every monomial. For every  $r \geq 1$  an explicit Hermitian matrix of size  $r + 1$  is strictly positive at every nonzero vector over  $\mathcal{P}_\Gamma$ , and even over its valuation closure, yet has inertia  $(1, r, 0)$  over  $\mathbb{C}((t^\Gamma))$ ; so the exponent group in the Hermitian spectral theorem of [surcomplex/spectral-theory](#) cannot in general be replaced by a finitely generated one. “Positive semidefinite” always means nonnegative at *every* vector over the full field. The engine is prime-denominator tails  $\sum_p t^{\delta(1-1/p)}$ : every finite-lattice approximant misses below valuation  $\delta$ , matching  $N$  terms algebraically needs degree exactly  $\prod_{j \leq N} p_j$ , and almost-disjoint prime sets give continuum many independent tails. A second theorem characterizes positivity of a two-scale matrix-valued measure  $M_0 + \varepsilon M_1$  by scalar null ideals and a vector-valued absolute-continuity condition — the two-exponent case of the matrix criterion that [surcomplex/hahn-herglotz-positivity](#) excludes; more exponents stay open. Transported to surreal normal forms. One manuscript; background inputs are named as such; not refereed, no Lean verification.

### 41. Automorphisms of the surcomplex field

[surcomplex/surcomplex-field-automorphisms/article.pdf](#)

from one manuscript, delivered 22 September 2026

“Automorphism of  $\mathbf{No}[i] = \mathbf{No}(i)$ ” has several inequivalent meanings, and this report keeps six apart: plain field, valued, value-fixing, 1-automorphisms, exponential, and those preserving the logarithmic modulus. The automorphisms fixing  $\mathbf{No}$  pointwise are the identity and conjugation; those preserving  $\mathbf{No}$  setwise form  $\text{Aut}(\mathbf{No}) \times C_2$ , a proper subgroup of all field automorphisms, as Hahn-series phase twists show by fixing every ordinary complex number and every value while moving the surreal real axis. Also the coefficient, character, value-group and leading-term-kernel layers of the valued automorphism group, automorphisms moving ordinary real constants by infinitesimals, zero-derivative field automorphisms and real forms. The exponential results — the amplification lemma, faithfulness, the negative answer to Kaplan–Krapp–Serra Question 5.4 and the logarithmic-modulus classification — are reprinted with attribution from [surreal/](#)

**exponential-automorphism-rigidity**; here  $\text{Aut}_L(\mathbf{No}[i])$  is observed to be a proper subgroup of the real-axis stabilizer. Readings of the Kaplan–Krapp–Serra paper beyond Question 5.4 are the source’s and were not rechecked. Not refereed; generic Lean coverage uses the rigidity report’s labels.

#### 42. The first- $\kappa$ coefficients: omitted types and completion in Hahn fields

[surcomplex/first-kappa-coefficients/article.pdf](#)

from surreal\_first\_kappa\_types.zip, delivered 22 September 2026

Inside a full Hahn field  $k((t^\Gamma))$ ,  $k = \mathbb{R}$  or  $\mathbb{C}$ , take the series with fewer than  $\kappa$  nonzero terms, for an uncountable and possibly singular cardinal  $\kappa$ . Which one-variable types does the full field realize that this subfield omits? Exactly those coded by the first  $\kappa$  nonzero terms of a series: these terms determine its complete ordered-field type in the real case and its valued-field type in the complex case, and the least number of formulas witnessing omission is  $\text{cf}(\kappa)$ , while in the pure complex field language all missing elements share one type. The same invariant gives the approximation cut, the least empty nest of closed balls, the valuation completion (the subfield is complete exactly when  $\text{cf}(\Gamma) \neq \text{cf}(\kappa)$ , and never spherically complete) and the valuation loss of linear retractions; Conway normal forms give surreal examples at  $\kappa = \aleph_\omega$ . The countable-support analogues in [surcomplex/three-duals-of-hahn-vector-spaces](#) and [surcomplex/rank-one-berkovich](#) are cross-referenced. The constructions and quantifier-elimination inputs are established; the proposed contribution is the joint classification. Not refereed, no Lean verification, limited priority audit.

#### 43. A single dilation recovers Hahn support

[surcomplex/single-dilation-hahn-support/article.pdf](#)

from surreal\_dilation\_support.zip, delivered 22 September 2026

Can one field automorphism make the Hahn structure of  $k((t^\Gamma))$  visible? For every coefficient-fixing monomial automorphism  $\sigma(t^g) = \chi(g)t^{\theta g}$  the report constructs a strong partial inverse of  $\sigma - \lambda$ , obstructed only at exponents fixed by  $\theta$  where  $\chi = \lambda$ , and for finitely many commuting ones a deformation retraction of the Koszul complex onto its joint-resonant coefficients, computing additive and principal-unit multiplicative cohomology. The main application: the single dilation  $S_2(\sum a_g t^g) = \sum a_g t^{2g}$  defines, in the language of fields with one automorphism, the coefficient field, every coefficient, the canonical monomials, the valuation ring and truncation. Hence its centralizer among *all* field automorphisms is exactly the untwisted coefficient–exponent lifts, full monadic second-order arithmetic is interpreted (so the theory is undecidable, with a  $\text{TP}_2$  witness), and all nonidentity rational dilations give the same definable data though no two are conjugate. The dilation is the  $S_a$  of [surcomplex/surcomplex-field-automorphisms](#), whose questions it does not answer. Elementwise results reach **No** and **No**[ $i$ ] through Conway normal forms, with proper-class qualifications stated. Not refereed, no Lean verification.

#### 44. Gamma and zeta over the surcomplex numbers

[surcomplex/gamma-and-zeta-functions/article.pdf](#)

from six manuscripts, delivered 22 September 2026

What can  $\Gamma(z)$  and  $\zeta(z)$  mean when  $z \in \mathbf{No}[i]$ ? Not one function: Taylor–Laurent evaluation at finite points, strong Hahn summation at infinity and the choice of a global exponential are kept apart throughout. At finite points the lifts have exactly the classical zeros and poles, so their Riemann hypothesis is the classical one, and stability of every finite zero under symmetric infinitesimal deformations is equivalent to RH with all zeros simple. At positive infinite real part every formal Dirichlet series is strongly summable, evaluation is an injective algebra map with exact Euler products and no zeros of zeta, every fibre  $\zeta = 1 + u$  is described explicitly, and zeta jets at dilations and at real shifts are algebraically independent over large fields (the prime lemma being Pong’s). Stirling, a phase-free ratio cocycle and Hurwitz zeta at an infinite parameter follow. With any global phase, Gamma, zeta and  $\xi$  exist on the whole horizontal tube; reflection then forces Gamma poles and zeta zeros on the phase’s infinite periods, and with the Ehrlich–Kaplan exponential zeta vanishes at the negative even omnific integers, while  $\xi$  keeps exactly the ordinary



zeros. At infinite imaginary part no exponential extending the finite-phase one gives reflection anywhere on the vertical strip, and resonant phases obstruct a coherent continuation; elementary extensions instead transfer classical zero-free regions, density bounds and de Bruijn–Newman results. *Merged from six manuscripts on the same questions; each was audited alone and all were compared claim by claim. No two contradict each other once domain, exponential, summation notion and deformation class are aligned; the reconciliation is printed at each point of use, and fifty-three minor source errors are fixed and listed.* Results the merge itself proves are marked. No proof of classical RH, no canonical Gamma or zeta on all of  $\mathbf{No}[i]$ , not refereed, no Lean verification.

### 3 Surquaternions

One report, which is where commutativity stops. It is filed on its own because commutative arguments from `docs/surcomplex` require review on moving to a division algebra: conjugation, multiplicativity of the modulus, valuation and leading coefficients, polynomial evaluation, matrix adjoints, exponential laws and every chain rule need their scalar and slice hypotheses checked. The report proves the additional noncommutative assertions with factor order retained.

#### 45. Surquaternions: algebra, infinitesimal geometry, and support-controlled analysis

`surquaternions/surquaternions/article.pdf`

from `surquaternions_article.zip` + `surquaternions_article(1).zip`

Hamilton’s quaternion division algebra over the surreals,  $\mathbb{H}_{\mathbf{No}} = \mathbf{No} + \mathbf{No}i + \mathbf{No}j + \mathbf{No}k$ , with central surreal coefficients, in three levels that are never silently mixed. *Finite algebra over an arbitrary real closed field*: division, conjugation, the field-valued modulus, complex slices and centralizers, conjugacy spheres, explicit square roots, the spin/rotation description with two independent surjectivity proofs, Cayley and stereographic presentations of one rational chart, quaternionic Möbius maps, one-sided polynomials with a constructive fundamental theorem and a complete zero-class algorithm, and finite-dimensional Hermitian spectral theory with SVD and polar decomposition. *Set-sized Hahn workspaces*  $D_{\Gamma} = \mathbb{H}(\mathbb{R}((t^{\Gamma})))$ : normal forms with a **noncommutative** residue algebra, exact multiplicative valuation and leading coefficients, the all-lengths Neumann support lemma proved through Higman’s finite-word lemma, the exact geometric inverse with its truncation certificate, the infinitesimal Hahn exponential and logarithm, BCH at positive valuation, the filtered nonabelian rotation group with its graded bracket, a spectral-projector estimate at arbitrary surreal gap scale with a sharp two-scale example, support-controlled implicit lifting, and an exact square-root conditioning analysis. *The full surreal class*: finite angles at arbitrary radius, the Ehrlich–Kaplan global phase normalization, the global radial exponential with its complete logarithm fibres and four-coordinate differential — rank-two critical spheres at positive radii in  $\pi\mathbf{Oz}$ ,  $\mathbf{Oz}$  the omnific integers, the infinite radius  $\omega$  among them — the componentwise Berarducci–Mantova derivation with a no-oscillation theorem and its chain-rule corollary, and coefficientwise slice, Cauchy and Fueter analysis on explicitly named coefficient spaces. **Section 2 is the ledger and should be read first**: three exponentials and four derivative-like operators carry overlapping names here, each satisfying a *different* law on a *different* domain, and the section fixes one name and one symbol per object and tabulates which satisfies which. It also explains why this report’s statement that the radial exponential is *not* an additive-to-multiplicative homomorphism is consistent with the physics report’s correct claim that the *commutative* Ehrlich–Kaplan exponential *is* one: the restriction to any single slice is exactly that commutative exponential and the law holds there, and it can fail across *noncommuting* arguments, where no common slice exists — witness  $\text{Exp}(\omega i + \pi j) = 1 + (\pi^2/2\omega)i + O(\omega^{-2})$  against  $\text{Exp}(\omega i)\text{Exp}(\pi j) = -1$ . A merged sentence blurring the two would manufacture a false contradiction across the collection. *Merged; roughly two thirds of the mathematics was common to both sources and is printed once.* Every `\label` in its source carries the prefix `squat:`. It makes no claim about any empirical subject.

## 4 Physics

One report, and the only one in the collection whose subject is not mathematics. It is filed on its own because its entire value is a firewall between what is proved and what is proposed, and because its conclusion is negative.

### 46. Surreal scalars in theoretical physics: black-hole singularities, asymptotic structure, and the limits of replacing the scalar field

[physics/surreal-scalars-and-spacetime/article.pdf](#)

from three manuscripts, delivered 21 September 2026

**The sentence this report exists to refuse, in any form, is that surreal numbers resolve the black-hole singularity.** Surreal and surcomplex scalars make the hierarchy of quantities near a gravitational singularity exactly representable; they do *not* make the spacetime regular. Four facts, each re-derived from the metric for the merge rather than copied from a source: the Kretschmann scalar of Schwarzschild is exactly  $K = 48M^2/r^6$ ; under a formal radial change  $r = s^q w(s)$ , with integer  $q \geq 1$ ,  $w(0) \neq 0$  and  $M \neq 0$ , and  $w \in \mathbb{R}[[s]]$ , the pole order becomes exactly  $6q$  with leading coefficient  $48M^2 w(0)^{-6}$ . The recorded checks use  $w(s) = 1 + s$ , whose constant term is one; this normalized case preserves  $48M^2$ . No substitution of the stated form removes the pole; the proper time from rest at  $R$  to  $r = 0$  is  $(\pi/2)\sqrt{R^3/2M}$ , which for *infinitesimal*  $R$  is infinitesimal and hence still finite, so the infall terminates either way; and the tidal component  $2M(2M - r)/r^4$  is unlimited at infinitesimal  $r$ . The exact identity  $r^6 K = 48m^2$  has *no* field-valued solution at  $r = 0$  in any characteristic-zero field, not merely in  $\mathbb{R}$ : evaluating at a nonzero infinitesimal radius names another point of a punctured domain and supplies no missing endpoint, no continuation law and no observable rule. What is true, and is what the report is for, is that the surreal/Hahn language records the power *and* the coefficient of every divergence exactly, keeps competing scales separate, and detects when an expansion has been used outside its regime — a gain in representation and nothing more. **The claim-status firewall** is the rest of its value: every theorem header and every substantive paragraph carries one of four visible tags — exact symbolic identity, checked exactly and never to a tolerance, which is the entire content of what the programs establish; conditional theorem, with hypotheses stated in full and *established for no physical model*, nothing in the tier machine-checked; assessment, meaning an argument and not a result; and imported, a provenance marker rather than a fourth tier. The article’s own tag counts are 53, 51, 140 and 34. *Merged; duplication between the three sources ran to about two thirds, all three independently rebuilding the static-spherical connection and the full Riemann tensor and re-deriving the same Schwarzschild quantities from the metric alone.* No priority claim; not refereed.

## 5 Foundations and computation

Five reports that take the rest of the collection, or the surreal field as a set-theoretic object, as their object. None belongs under **surreal** or **surcomplex**: the first three range over the whole subject, and the last two are set theory about **No** rather than algebra or analysis inside it. Each of the first three is a merge of three manuscripts delivered together, the first two on 21 September 2026, and in those two the hazard was not overlap but a collision of near-identical names. The last two are merges of four and five manuscripts delivered on 22 September 2026, each proving one spine several times under different hypotheses.

## 47. Foundations for surreal and surcomplex mathematics

[foundations-and-computation/foundations/article.pdf](#)

from `surreal_surcomplex_foundations.zip` + `surcomplex_foundations.zip` + `surreal_foundations.zip`

In what foundation does this mathematics legally live, and how would a proof assistant encode it? Compares ZF/ZFC with definable classes, GB/GBC/NBG, Kelley–Morse, Grothendieck universes and TG, Feferman reflection, constructive set theory, dependent type theory, univalent foundations and native Lean, and proves the size obstructions each must respect: no unrestricted cut operation can accept every subclass of its own carrier, the surreals are not Dedekind complete, and a universe cannot cut itself. The positive counterpart is a localization theorem — every *set* of surcomplex numbers already lies in one divisible set-sized workspace  $K_\Gamma = \mathbb{C}((t^\Gamma))$  — with a finite-algebra descent theorem explaining why a finite zero scheme gains no points in a larger workspace. Carries two series over  $\mathbb{C}((t^\mathbb{Q}))$  that differ only in the sign of an exponent and reach opposite conclusions:  $\sum t^{-n^2} X^n$  is summable at no nonzero argument, while  $\sum t^{+n^2} z^n$  is summable everywhere and is a *nonpolynomial* function coherent at every radius — which does not refute the all-scale rigidity theorem of `surcomplex/analysis`, stated for class functions on the whole `No[i]`, but shows it does not transfer to a fixed value group. *Merged; the danger was aggregation. All three manuscripts read the same Lean repository at the same revision on the same day, so their agreement is one act of source reading performed three times, not three confirmations. The findings are printed once and the verification boundary has its own early section.* The original merge compiled three illustrative Lean files and inspected their axiom dependencies. Those historical checks are distinct from the later development mapped in `FORMALIZATION.md`. The maintained article also makes the universe-relative localization hypotheses explicit, expands the positive-support recursion proof, and separates intrinsic convergence from full fine topology.

## 48. Surreal and surcomplex numbers in symbolic computer algebra

[foundations-and-computation/computer-algebra/article.pdf](#)

from `surreal_surcomplex_cas.zip` and two same-named archives

What can actually be computed exactly, and what only approximated or denoted? Separates finite exact denotation, effective coefficient access, decidable equality and order, and certified approximation into capability tiers, then works through an ordered rational-monomial core, real-algebraic closure, certified Hahn series, formal germs, analytic lifts, transseries, Newton profiles, Hensel lifting, finite quotient algebras, residues, and a proposed Wolfram Language architecture. *Merged; the danger was three prototypes with one name.* Each manuscript ships an executable Wolfram package, two of them near-anagrams of each other, two exporting six identical public symbol names with incompatible signatures, and all three calling themselves the rational monomial core. They are three different objects: coefficients  $\mathbb{Q}$  in one and  $\mathbb{Q}(i)$  in the others; exponent lattices  $\mathbb{Z}^d$  against  $\mathbb{Q}^r$ , so a ramified root is representable in one and not the next; and two of them order their monomial variables by opposite conventions, requiring the corresponding translation of monomial-order inequalities. The report never writes “the prototype” and tabulates the divergences instead. All three check suites were rerun for this repository and reproduce exactly — 34/34, 110/110 and 88/88 in Wolfram, 69/69 and 323/323 in Python — but one package cannot be loaded at all as shipped: an unclosed bracket stops it parsing, which is why its own record submits definitions rather than loading the file.

## 49. Computable surreal numbers, merged and reconciled

[foundations-and-computation/computable-surreals/article.pdf](#)

from `computable_surreal_numbers.zip` + two same-named archives

Which surreal numbers admit an algorithm, and in which representation? Three manuscripts arrived with two different numerical representations of the same objects — one using effective rational left-finite series with finite candidate covers, two using computable bounded-denominator Puiseux series — and the report is one integrated development rather than a concatenation: it preserves the distinct representations instead of picking one, supplies an explicit reconciliation ledger disposing of every divergence, keeps complementary proofs where the two routes prove the

same statement by different means, and separates mathematical existence from the existence of a uniform algorithm throughout. *Merged; and this is the one report whose source manuscripts were never redistributed with it in the first place.* The ledger cites them by their original  $\text{\TeX}$  line numbers and `data/provenance-manifest.json` records their archive identities, file hashes and line counts, so every disposition is traceable against a copy obtained elsewhere. Four verification suites are rerun and pass — 4,967 exact comparisons, 18 named unit tests, 266 exact checks, and 4,366 exact cross-model and cover-contract checks in the updated reconciliation suite. The latter compare dense common-grid Puiseux convolution against sparse rational convolution, candidate covers with extra zeros and nonnested lists (including negative zero candidates in the geometric-series construction), the finite-jet root transformation, derivative and primitive identities, and finite instances of the two unbounded-denominator support examples. *The four units mean different things and are deliberately not added together.* A finite test proves no infinite theorem, no real closedness, no computability classification and no undecidability reduction; the synthetic halting fixtures in the original programs do not compute the halting set; the package implements neither arbitrary computable reals nor arbitrary infinite surreal arithmetic; and there is no Lean-checked formalization and no literature-wide priority claim.

## 50. Surreal fields across set-theoretic universes

[foundations-and-computation/surreal-fields-across-universes/article.pdf](#)

from four manuscripts, delivered 22 September 2026

How does the surreal field of an inner model look from a larger universe with the same ordinals? For transitive  $M \subseteq N$ , the old field  $\mathbf{No}^M$  is an elementary subfield of  $\mathbf{No}^N$ , yet its saturation, counted in  $N$ , can fail: for every uncountable cardinal  $\kappa$  of  $N$  it is externally  $\kappa$ -saturated exactly when  $N$  adds no ordinal-valued sequence of length below  $\kappa$ , and one small side of a cut already suffices; at  $\omega$  the criterion is instead that no reals are added. A shortest new sequence, of regular length  $\delta$ , yields an explicitly decoded omitted cut of character  $(\delta, \delta)$ , and every externally set-presented gap is the trace of a unique fresh sign sequence all of whose proper prefixes are old. The first new birthday  $\beta$  satisfies  $\delta \leq \text{cf } \beta \leq \beta$ , and Cohen, closed and Prikry forcings are computed. The pure field  $\mathbf{No}^M[i]$  stays saturated over every set of parameters and, under Global Choice, is isomorphic to  $\mathbf{No}^N[i]$ ; naming conjugation restores the real spectrum, which gives real forms of one algebraically closed class field that are not conjugate to the standard one and have no set-sized cofinal subset. Outer sets of old numbers remain closed and uniformly discrete, and a set-complete class Boolean algebra in which reverse simplicity is dense answers a question from Berenbeim’s 2020 notes. *Merged from four manuscripts proving one spine; the interval-tree and sign-block-code constructions are kept as second routes, and results under plain ZFC are kept apart from those using Global Choice.* The real-form results continue, but do not settle, the real-form question of `surcomplex/surcomplex-field-automorphisms`. Three finite suites check sign codes and interval codes; a finite check proves no infinite theorem. Not refereed, no Lean verification, priority not certified.

## 51. Birthday cutoffs and hereditary sets

[foundations-and-computation/birthday-cutoffs-and-hereditary-sets/article.pdf](#)

from five manuscripts, delivered 22 September 2026

Which set universe does a bounded piece of surreal arithmetic remember? The ordered field is real closed and sees nothing, but once birthday is named, the surreals born below an epsilon number  $\lambda$  interpret the hereditary sets  $H_{\kappa(\lambda)}$ , with  $\kappa(\lambda)$  the least cardinal not below  $\lambda$ , and are bi-interpretable with  $(H_{\kappa(\lambda)}, \in, \lambda)$ ; already  $\mathbf{No}_{<\varepsilon_0}$  recovers every hereditarily countable set. Sign bits are read from order and birthday, a bounded ordinal pairing packs relations, and small certificates for Conway arithmetic avoid assuming Replacement in  $H_\kappa$ . Consequences: an exact classification of elementary inclusions between cutoffs, sentences for Power Set, Replacement and worldly cutoffs, the exact range of the coding without Choice, new-subset detectors and the first new birthday of an outer model, a Prikry extension that leaves a birthday-enriched field unchanged while singularizing its cutoff (from a measurable cardinal), and, for surcomplex cutoffs

with conjugation and a coordinate birthday, the orientation obstruction and two lifts of every real elementary embedding. The full-class bi-interpretation with the set universe is the theorem announced by Chen–Hamkins–Yang, recorded in **foundations-and-computation/foundations**; it is printed with a written proof as that announced theorem, not as a new result. *Merged from five manuscripts proving one spine at different cutoff classes; the most general is the base, and two arithmetic certificates and two set codings are kept as different proofs.* Five finite suites pass; a finite check proves no infinite theorem. Not refereed, no Lean verification, priority not certified.